# KOCHI METRO RAIL LIMITED

# Request For Proposal

**BID ID: GEM/24/B/5728971**

Supply, Installation, Commissioning, Training and CAMC of

X- Ray Baggage Scanners & Door Frame Metal Detectors

KOCHI METRO RAIL LTD.

4th Floor, Jawaharlal Nehru Stadium Metro Station,

Banerji Road, Kaloor, Kochi – 682017

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# CHAPTER -1
# DEFINITIONS AND ABBREVIATIONS

In this tender document and associated documentation, the following terms shall be interpreted as indicated below:

1. "Services" means services to be provided as per the requirement mentioned in the scope of work
2. KMRL means Kochi Metro Rail limited
3. “Noncompliance” means failure/refusal to comply the terms and Conditions of the tender;
4. “Non responsive” means failure to furnish complete information in a given format and manner required as per the tender documents or non-submission of tender offer in given Forms / Pro forma or not following procedure mentioned in this tender or any of required details or documents is missing or not clear or not submitted in the prescribed format or non-submission of tender fee or EMD declaration form.
5. Bid means the proposal submitted by the Bidder(s) in response to this Tender in accordance with the provisions hereof. Wherever „Tender‟ word is used, it shall mean the same as “Bid”.
6. Bidder means an applicant who has submitted the Bid as per notice inviting tender of this Tender document.
7. Bid Security or “Earnest Money Deposit” or “EMD” shall have the meaning ascribed to it in hereof.
8. Bid Process means the process of selection of the Service Provider through competitive bidding and includes release of this Tender document, submission of Bids, scrutiny and evaluation of such Bids as set forth in the Tender.
9. Effective Date shall mean the date on which the work order is issued by KMRL.
10. Letter of Award or “LOA” means the letter or memorandum communicating to the Successful Bidder the acceptance of its Bid.
11. Tender Document or “Tender” shall mean this document and shall include any addendum(s) issued thereto.
12. Successful Bidder shall mean the Bidder, whose Bid is evaluated as the most competitive following the bid evaluation process as set forth in this Tender Document and to whom work order is issued subsequently.
13. Incoterms means the trade terms published by International chamber of commerce (ICC)

14. Similar Items means Supply, installation, testing & commissioning of X-Ray Baggage Scanners and Providing satisfactory DLP/CAMC support.
15. XBIS means X-RAY Baggage Inspection System or X-Ray Baggage Scanner
16. DFMD means Door Frame Metal Detector.
17. DLP means Defect Liability Period.
18. CAMC means Comprehensive Annual Maintenance Contract.

# CHAPTER -2

## GENERAL INSTRUCTIONS OF THE TENDER

### SECTION 1 - PREBID

1. The Tender and any order resulting from this Tender shall be governed by our conditions of contract and the vendor quoting against this Tender shall be deemed to have read and understood the same.

2. Where the tenderer has offered Counter Terms and Conditions of business, we shall not be deemed to be governed by those unless our specific written acceptance thereof has been given. Counter Terms and Conditions may be mentioned along with format for deviations as per Appendix 4. Unless the counter terms/deviations are specified in the prescribed format it is deemed that all the terms and conditions including technical specifications are acceptable to the bidder and the offer has been submitted accordingly.

3. No Conditions and Terms, notice of which has not been given by the tenderer while submitting tenders, will be considered by us, if put forward in subsequent correspondence.

4. **TENDER**

   a) Bid and all other related documents are to be submitted in English. Supporting documents, which is in other language, should be translated in English and submitted along with copies of original.

   b) The intending Bidders must read the RFP carefully and should only submit his Bid if he considers himself eligible and he is in possession of all the documents required.

   c) Offers directly from the manufacturers of the equipment are desirable. Where authorized dealers or agents of the manufacturers are quoting on behalf of their principal, they should submit a tender specific letter of authority from the Principals. Offers of brokers and middlemen will not be accepted. One agent or dealer cannot represent more than one manufacturers and cannot quote on their behalf in a particular tender case.

   d) If Authorized dealer is participating in tender then Furnish the tender specific authorization letter from manufacturers, to submit offer against this tender.

e) If the Tenderer is a firm or company, the tender documents to be submitted by the authorized signatory of the firm/company.
f) The Bid shall be prepared and shall be signed in all pages by the Bidder or a person or persons duly authorized to bind the Bidder to the Contract.

g) Tender not submitted in the appropriate forms is likely to be rejected.
h) On the due date of opening, the tender will be opened online.
i) Bidders black listed/ordered for blacklisting by Govt. of Kerala/India or KMRL cannot be considered.
j) KMRL does not bind itself to accept the lowest tender or to assign any reason for non-acceptance of the same.
k) Landed cost shall be considered for price comparison and order placement.
l) The manufacturer/ Authorized dealer/Contractor/supplier participating in the Tender hereafter will be called as "Applicants".

m) All costs for site visits, obtaining information/ data and preparation/ meetings, etc. in relation with Bid submission shall be borne by the applicants.

n) KMRL reserves the right to verify all the credentials and inspect their executed orders etc. to satisfy themselves about their performance and capability to execute the supply satisfactorily, if found any failure on submitted details, KMRL holds the right to invoke performance security or Bid security wherever applicable

o) KMRL reserves the right to not to choose any applicant, relinquish the Bid process or reissue Bid with or without modification as it chooses. KMRL is not bound to furnish any explanation as to its decision to any of the participants.
p) Further, corrigendum/ addendum, if any, issued to the Tender /bid document, shall only be made available at GeM portal.

q) The Bid of all Bidders who have submitted EMD/Relevant Exemption certificate with bid security declaration only will be opened.

r) If such nominated date for opening of Bid is subsequently declared as a Public Holiday by the Employer, the next official working day shall be deemed as the date of opening of Bid

s) The quoted line items shall be in accordance with the Bureau of Indian Standards as per MoHUA notification No.V-11/15/2020-BIS dated 01.02.2021 (if applicable)

t) Public Procurement (Preference to Make in India) as per Ministry of Commerce & Industry DPIIT notification No. P-45021/2/2017-PP(BE-II) dated 29.05.2019 and subsequent amendment will be applicable.

## 5. LOCAL CONDITIONS

It will be imperative on each tenderer to fully acquaint himself of all the local conditions and factors, which would have any effect on the performance of the contract and cost towards the supply covered in this tender. The Employer shall not entertain any request for

clarifications from the tenderer regarding such local conditions. No request for the change of price, or time schedule of delivery of machines, shall be entertained after the Employer accepts the offer.

6. **CLARIFICATIONS**

Please note that queries related to enquiry specifications, terms & conditions etc. should be submitted 07 days before the bid submission end date of the tender through GeM portal only. M/s. Kochi Metro Rail Ltd (KMRL), at its sole discretion, may not entertain the queries sent by post/fax/emails or through any other mode of communication. The KMRL will publish the reply to all such queries as corrigendum if required.

7. **EMD (EARNEST MONEY DEPOSIT)**

a) **Forms of EMD**

Bidders can also submit the EMD with Account Payee Demand Draft in favour of KOCHI METRO RAIL LIMITED payable at KOCHI

Bidder has to upload scanned copy / proof of the DD along with bid and has to ensure delivery of hardcopy to the Buyer within 5 days of Bid End date / Bid Opening date.

Bidders can also submit the EMD with Banker's Cheque in favour of KOCHI METRO RAIL LIMITED payable at KOCHI

Bidder has to upload scanned copy / proof of the BC along with bid and has to ensure delivery of hardcopy to the Buyer within 5 days of Bid End date / Bid Opening date.

Bidders can also submit the EMD with Payment online through RTGS / internet banking in Beneficiary name. Kochi Metro Rail Limited
Account No. 0803201002929
IFSC Code: CNRB0000803
Bank Name: Canara Bank
Branch address: Ernakulam Broadway Branch, Ernakulam Kerala 682031

EMD can also submitted through BG as per the format provided in Appendix 12.

**<u>Payment of EMD/Bid Security:</u>**

a. The Bidder can submit the EMD of Rs. **6,02,256 /-** in the form of Payment online through RTGS /internet banking .On-line payment shall be in

Beneficiary name Kochi Metro Rail Limited.
Account No : 0803201002929
IFSC Code : CNRB0000803
Bank Name : Canara Bank
Branch address: Ernakulam Broadway Branch, Ernakulam Kerala 682031

b. Bidder has to upload scanned copy / proof of the Online Payment Transfer in place of EMD along with the bid document

c. The EMD of the successful bidder shall be returned upon the acceptance of the Contract Order and on the receipt by the Employer of the Performance Security in accordance with clause mentioned in the Buyer Added Bid Specific Terms and Conditions.

d. The Bid security (EMD) remitted by the unsuccessful bidders shall be released after placing of Contract Order to the successful bidder or finalization of contract.

e. Bid security shall be exempted as per GeM GTC. In case EMD exemption required, valid exemption certificates along with Bid security declaration as per Appendix 9 to be submitted along with the technical bid.

f. Bids received without EMD/Valid exemption certificate along with Bid security declaration shall be prima-facie rejected from further evaluation. Bidders are requested to provide duly notarized tender security declaration as per Appendix-9.

g. Bid Security submitted by the bidder shall be forfeited, if the bidder:

i. Withdraws or modify or impairs or derogates from the bid in any respect within the period of validity of its bid; or
ii. If it comes to notice that the information / documents furnished in its bid is false, misleading or forged; or
iii. Fails to furnish requisite performance security within stipulated time required a s per e-bid / RA conditions.

8. **BID VALIDITY**

The bid shall be valid for a period of minimum 120 days from the date of closing of the bid and to be extended as per the request from KMRL

9. **COMPLIANCE OF REQUIREMENTS**

a) The items offered should be in accordance with the stipulated description, and specifications in "CHAPTER 3-TECHNICAL SPECIFICATION". Compliance with the specifications should be clearly indicated in Appendix 3.

b) Whenever if the tenderer deviates from the provisions of a clause/sub- clause, they shall furnish detailed justification for the same in the Statement of Deviations in the format given in Appendix 4

10. **TERMS OF PRICE**

a. The tenderers should quote their lowest possible price on DDP KMRL basis (Incoterms 2020)
b. The bidders should quote the price for door delivery inclusive of GST, all other applicable taxes, duties, Transit insurance, freight charges, loading & unloading charges, arrangement/cost of machines, Equipments, labour, tools, materials, consumables, supervising, commissioning and all other related charges required for the execution of entire contract. If GST exemption is there, appropriate exemption notification document should

be submitted. Transportation, packing, forwarding, installation, dismantling, disposal and other charges if any, shall also be included in the quoted price

c. Bidders are advised to check applicable GST on their own before quoting. Buyer will not take any responsibility in this regards. GST reimbursement will be as per actuals or as per applicable rates (whichever is lower), subject to the maximum of quoted GST %.
d. Contract price will be the total price quoted inclusive of all taxes and including GST.
e. The tax liability as per GST Act rests with the contractor, ambiguity if any regarding the percentage for that particular work or Services Accounting Code (SAC)/HSN etc shall be cleared before offering of rates.
f. Any liability on account of wrong classification of goods or services for arriving at taxes will be on the bidder.
g. The prices quoted should be firm and not subject to any variation, unless specified in the tender documents. Any deviation in this aspect may make the offer liable to be ignored.
h. Invoice as per GST Act should be furnished prior to every running account/final bills.
i. The GSTIN of Kochi Metro Rail Ltd is 32AAECK5274H1ZL.
j. GST at actuals shall be paid on the quoted/corrected base value as per GST law in India after submission of appropriate GST invoices with GSTIN of the bidder and GSTIN of Kochi Metro Rail Ltd - 32AAECK5274H1ZL.
k. Under GST law the contractor shall also comply regarding filing of all the returns to the GST network/government departments within the stipulated time every month or such other period as required by the Government. If the contractor does not comply with any of the GST laws and procedures and if KMRL incurs any liability on this account or does not get the input credit from the GST Network/Government as goods and/or service receiver due to the contractor's failure to comply with the procedures of filing / uploading of data/submissions of documents etc in time then all such liability including the input credit of the GST lost by KMRL and the penalties and interest incurred by KMRL would be the liability of the contractor and the same shall be recovered either by recovery from security deposits / any other amount payable by the contractor to KMRL or through direct payment. The contractor shall submit the copy of latest filed return - GSTR1 along with the invoice

## 11. POWER OF ATTORNEY

Only authorized person shall sign the tender document on behalf of the tenderer. Proof of authorization in the form of Power of attorney (sample format in Appendix 2) /Board Resolution or any other form such as inclusion of the name in the license document or any other legal document shall be submitted by the tenderer.

## 12. DOCUMENTS TO BE FURNISHED ALONG WITH TECHNICAL BID

The following documents to be submitted along with the technical bid.

a. Detailed technical description of each of the equipment offered
b. Details regarding power supply equipment such as UPS, invertor .
c. The documents complying to general requirements as per clause 1.3, 2.3 and certifications of chapter 3 of this RFP (National/International Standards/Regulations

for safety and EMC requirements, ISO 9001, CE, AERB, International Radiation Safety Standard, ISO1600/33 etc..)

## 13. INDIAN REPRESENTATIVE & HIS SERVICES/ FACILITIES IN INDIA

13.1. The foreign tenderer shall include in his offer the name of the person /firm who will be acting as his representative in India in respect of his offer. He shall also indicate the after sales service facilities through his representative in India.

13.2. Foreign firms quoting directly against the inquiry through their representatives must specify the servicing facilities in India.

13.3. Tenderers of foreign firms should furnish following particulars.

i. The name and address of the Indian representative.

ii. The precise relationship between the foreign manufacturer /principals and their Indian representative

iii. The mutual interest which the manufacturer /principal and the Indian representative have in the business of each other

iv. Foreign Tenderer has to submit a certificate that the bidder is not having any Commission Agent in India and no agency commission will be paid, otherwise it shall be sufficient ground for rejecting of his offer. Indian representative should also mention Income tax permanent account number and certification of registration of OEM and registration of Indian representative.

v. All services namely supply installation, commissioning, training and maintenance including DLP & CAMC need to be rendered by the Indian representative.

vi. Certificates /documents indicating for Past performance of last 10 years.

## SECTION 2 - BID EVALUATION

### 14. MINIMUM ELIGIBILITY CRITERIA

The bidder/applicant should submit the following documents as a minimum eligibility criteria documents. The bids will not be considered for evaluation if any of the documents pertaining to minimum eligibility criteria are not submitted along with bid.

| Clause no | Description | Documents |
|---|---|---|
| 14.a | Bidder should have successfully carried out the following;<br><br>a) Supply, installation, testing & commissioning of X-Ray Baggage Scanners<br><br>AND<br><br>b) Provided satisfactory DLP/CAMC support to X-Ray Baggage Scanners which is not less than for a period of 5 years<br><br>AND<br><br>c) Supply, installation, testing & commissioning of Door Frame Metal Detectors. | The proof should be submitted in terms of<br><br>The bidder shall submit GeM contract order/CA/Notary certified Purchase order (s) for X-Ray Baggage Scanners as follows;<br><br>● At least one order of value not less Rs.4.81 crore<br>**OR**<br>● Two orders each of value not less than Rs.3.01 crore<br>**OR**<br>● Three orders each of value not less than Rs.2.40 crore.<br><br>Issued by Metro Rails / Airports / Central / State Government Organizations / PSUs for supplying X-Ray Baggage Scanners along with DLP, CAMC support during last 10 years ending 31.12.2024 with (a) AND (b) as follows;<br><br>a) CRAC/ Acceptance certificate/ Delivery/work completion certificate with value of supply/work done along with relevant Contract/Purchase order from the organization/consignee issued for the supply, installation & commissioning.<br><br>AND<br><br>b) CRAC/Certificate of satisfactory performance of DLP/CAMC for a period of minimum 5 years for the work in a) from the organization/consignee to whom the delivery is made. |

| | | |
|---|---|---|
| | | AND<br><br>c) The bidder shall submit GeM contract order/CA/Notary certified Purchase order of atleast one completed work of Door Frame Metal Detectors in the preceding 3 years ending 31.12.2024 with CRAC/ Acceptance certificate/ Delivery/work completion certificate with value of supply/work done from the organization/consignee issued for the supply, installation & commissioning.<br><br>**it may please be noted that the order/s submitted by the bidder as a proof of experience also can be any ongoing order/s meeting the above criteria for** X-Ray **Baggage Scanners.** |

| | | |
|---|---|---|
| 14.b | Tenderer shall be eligible fulfill minimum Eligibility criteria of having an average annual turnover for last three financial years should not be less than Rs.1.80 crore. | Financial data for last three financial years ie. CA audited balance sheet, Profit and loss statement for financial year 2021-22, 2022-23 and 2023-24 has to be uploaded by the tenderer in Appendix-11 of RFP. Chartered Accountant shall certify the financial data in the prescribed format with his stamp and signature. In case audited balance sheet of the last financial year (2023-24) is not made available by the bidder, the bidder has to submit an affidavit (As per format provided in Appendix-11 A of this RFP) certifying that 'the balance sheet has actually not been audited so far'. In such a case the financial data of all the financial years will be taken into consideration for evaluation. However, if the affidavit is not submitted as per the format mentioned above, the turn over for that particular year will be considered as "zero" for the calculation of average annual turnover. |
| 14.c | Compliance with all technical parameters and scope of work as per Chapter 3 of RFP is required. | Schedule of Compliance with technical specification of items as per Appendix 3 of this tender document to be submitted. |
| 14.d | An undertaking to the effect to be given by the bidder that they have not been blacklisted /debarred by any government/government undertaking/ semi-government/ Govt. controlled institutions/ reputed private organizations in India or Abroad. | An undertaking as mentioned in Appendix 8 of this tender document to be submitted. |
| 14.e | The Principal/ licensed Manufacturer of X-RAY Baggage Screening System; or its authorized representatives in India are only eligible to quote for the tender. | Manufacturing & supply license of manufacturer. In addition, for authorized representatives of manufacturer - A letter from the manufacturers (Memorandum of Understanding – MOU) shall be required and to be submitted along with tender document where in manufacturer has authorized the bidder to participate in this tender on their behalf or letter authorizing the firm on sales & service of equipment by manufacturer. |

| | | |
|---|---|---|
| 14.f | 1. Original Equipment Manufacturers of X-RAY Baggage Screening System or its authorized representative must have office in India.<br>2. The principal manufacturer or its authorized representative in India should have their own service centre.<br>3. For the smooth maintenance and to reduce the downtime during DLP & CAMC of Security systems, it is mandatory that the bidder should have a local office or a maintenance partner in Kochi. If not, the bidder must have to start a local office or engage a maintenance partner in Kochi. Proof in this respect to be submitted along with the bid. | The address and contact details of office & service centre in India. Details of local office in Kochi shall be submitted for verification.<br><br>If the bidder is not having a local office in Kochi, the bidder must have to start a local office or engage a maintenance partner in Kochi within 01 month from the date of PO/contract and the declaration for the same has to be submitted (with seal & sign) along with the technical bid. |
| 14.g | In case of any termination or dispute of contract /agreement /relationship between Indian representative and OEM (Original Equipment Manufacturer), OEM will be laid with full responsibility of carrying the DLP along with CAMC work as mentioned in the tender document. | An undertaking for the same from the OEM is required to be submitted in the technical bid. |

## 15. DETERMINATION OF RESPONSIVENESS

Prior to evaluation of Bid, the KMRL will determine whether each Proposal is responsive to the requirements of the Tender. The KMRL may, in its sole discretion, reject any Proposal that is not responsive hereunder. A Proposal shall be considered responsive only if:

15.1. It is received by the Proposal Due Date including any extension thereof

15.2. It is accompanied by the EMD/valid exemption certificate with duly notarized EMD declaration as per Appendix 9.

15.3. Applicants has submitted all the required documents in format as mentioned in the Tender

15.4. Has submitted an offer for all required items.

## 16. AWARD CRITERIA

a. The Bid of the bidder whose Bid is responsive, complete and in accordance with the Tender Document will be considered for evaluation. After completing the evaluation, GeM Contract/PO will be placed to the technically qualified least total cost including all taxes of the offers received.

b. The successful bidder shall provide order acceptance in GeM within 7 days from the date of receipt of the same.

## 17. RIGHT TO ACCEPT ANY BID AND TO REJECT ANY OR ALL BIDS

Notwithstanding anything contained in the Tender Document, KMRL reserves the right to accept or reject any Bid in its sole discretion or to reject all Bids and the bidding process without assigning any reason whatsoever at its sole discretion at any time before issuance of a Letter of Acceptance without incurring any liability.

# CHAPTER -3
# TECHNICAL SPECIFICATION & SCOPE OF WORK

## A. TECHNICAL SPECIFICATIONS:

### 1. X-RAY BAGGAGE INSPECTION SYSTEM :

#### 1.1. GENERAL:

X-RAY Baggage Screening System shall be located at the specified location of KMRL metro stations or other places as per site requirement to screen passenger's baggage before permitting entry to platforms and vital locations of KMRL premises.

SCOPE:

a) The specification of X Ray Baggage Screening System covers technical specification and requirement of baggage screening systems consisting of X Ray machine, entry/exit stainless steel roller tables, Operator control console, Table and chair for operating the operator console, UPS, LCD color monitors, etc. for baggage screening at different locations of Metro stations.

b) The machine should be capable of storing images and these images can be viewed and retrieved through a USB port using an external drive. The system shall be able to work continuously without interruption.

#### 1.2. SYSTEM DESCRIPTION:

a) X Ray Baggage Screening System to be provided at Metro Stations of KMRL shall consist of X Ray machine, Operator console with suitable controls, stainless steel roller tables on entry and exit side, online UPS with maintenance free batteries for giving 30 minutes power back-up, LCD colour monitors, Power cables, Combined Test Piece (CTP), Threat Image Projection (TIP) software, stabilizers, etc.

b) X Ray Baggage Screening System shall have tunnel size of 600 (+8%) mm (width) x 400 (+8%) mm (height)

c) The X ray baggage screening system should enable the operator to perform all related tasks for screening & monitoring of objects.

d) The X ray baggage Operator console shall be an independent unit (not integrated with the X Ray Baggage Scanner unit).

## 1.3. **GENERAL REQUIREMENTS:**

a) The equipment shall adopt widely accepted technology for its operation and shall be of proven performance. The equipment shall conform to international & national standards/regulations for safety & EMC requirements.

b) Original equipment Manufacturer or their authorized representative of X Ray Baggage Screening System shall have their service based in India.

c) X Ray Baggage Screening System shall have quality system compliance (ISO 9001), CE complied and (AERB) Atomic Energy Regulatory Board of India certified. International Radiation Safety Standards like that of 21-CFR1020.40 of USA or equivalent is desirable. Manufacturers will have to produce a certificate in this regard for the particular model proposed in the bid.

d) The X-Ray machine shall comply with the film safety of ten passes of ISO1600/33 DIN high-speed photographic film.

e) All software and firmware upgrades shall be free of cost for a period of 10 year during (DLP plus CAMC). The PC of the machine shall be easily accessible and programmable.

f) The system should be capable of adapting to diverse operational requirements and evolving security challenges through easy customization of software-based controls and tools.

g) The power supply available at the Metro Station shall be 230V, 50Hz AC which may vary from 160 Volts to 270 Volts. All modules of the Screening System should work using this power supply only with requisite converters and voltage stabilizers of suitable capacity (Minimum of 2 KVA capacity).

h) The X Ray Baggage Screening System shall be capable of bidirectional screening operation.

i) Anti-rodent and dust proof cover shall be provided for the system.

## 1.4. **TECHNICAL REQUIREMENTS**:

a) X Ray Baggage Screening System for all type of Metro stations should mainly consist of the following:

i. X-Ray system.
ii. Operator Control Console.
iii. Lockable drawer for console/screen monitor.
iv. Entry/ Exit Roller Table.
v. UPS and Power Stabilizer (Minimum of 2KVA or above from reputed supplier).
vi. Battery and Stabilizer Stand.

b) The X-Ray machines should preferably be provided at specified location of the metro station or any other locations as decided by the purchaser in KOCHI, Kerala.

c) The X Ray Baggage Screening System shall have imaging capabilities using dual energy folded detector array having minimum number of diodes for following tunnel sizes:
For tunnel size 600 (+8%) mm (width) x 400 (+8%) mm (height): Sensors > 1000 photo diodes.

d) The X Ray Baggage Screening System shall have transparent color feature providing a crisp, clear and robust color images enabling the operator to interpret threat objects with higher degree of confidence.

e) The X Ray Baggage Screening System shall have an operator interface with a touch sensitive pad /roller allowing operator for a continuous heads-up operation.

f) The machine should be capable of discrimination of organic /inorganic stripping, reverse video, pseudo colour imaging, image annotation, real time continuous diagnostics and edge enhancement.

g) Continuous Electronic Zoom facility should be available to magnify the chosen area of an image eight times (8X) or more. Image features shall be keyboard controllable.

h) Each machine should have Threat Image Projection (TIP) software which can insert fictitious threat image at periodic intervals and tracks operator responses, allowing supervisor to access the effectiveness of the screening operations and encouraging operator to stay alert.

i) The system shall have image archive option that supports storage of minimum 10,000 images with date & time stamping, determined by the user defined expiration dates or on a first-in, first out basis.

j) The system shall have audio and visual alarm for a threat after it has compared scanned objects to data on known threats and shall highlight suspect items for further analysis.

k) All the equipment's installed shall be able to withstand operating temperature $0^0$ C to $40^0$C and humidity of 95 % non-condensing. The storage temperature capacity should be -$20^0$C to $60^0$C and airborne noise level of <70 dB (A).

1.5. **HARDWARE REQUIREMENT:**

I. X-Ray System of Tunnel Size: 600 (+8%) mm (width) x 400 (+8%) mm (height)

II. The machine should work on 230 V AC voltages, single phase, 50 Hz, 15 Amp max. The power supply available at the Metro Station shall be 230V/ 50Hz AC which may vary from 160 Volts to 270 Volts. The equipment should be able to with stand the voltage fluctuations.

III. The X-Ray machine shall meet the following minimum technical requirements.

| | |
|---|---|
| Conveyor Height | 600mm – 800mm from floor |
| Conveyor Speed | 200 mm per second (minimum) or above |
| Conveyor Capacity | 150 Kg (minimum) or above |

IV. X-Ray Requirements: The X-Ray machine shall have following minimum technical specification.

| | | |
|---|---|---|
| i. | X-ray Voltage | Maximum 160 KV |
| ii. | Duty Cycle | 100%, No warm-up procedure |
| iii. | Cooling | Sealed Oil Bath / Sealed Di-electric bath with forced air |
| iv. | Beam Orientation | The X-ray beam should be such that the complete image at maximum size of bag is displayed without corner cuts. |
| v. | X Ray Sensor | Sensors > 1000 diodes, L-shaped detector (Folded array type) configuration. In case of defective diode arrays, scanning should be disabled and error message should be displayed on the screen. |
| vi. | Radiation level | The radiation level should not exceed accepted health standard (0.1mR/Hr. at a distance of 5 cm from external housing). Relevant certificate from AERB required. |

| | | |
|---|---|---|
| vii. | Storage Temperature | -20$^0$C to 60$^0$C. |
| viii. | Operating Temperature | 0$^0$C to 50$^0$C. |
| ix. | Humidity | 95% non-condensing |

V. X-Ray Imaging and performance: The X Ray machine shall have the following minimum technical specifications and features for imaging:

| | | |
|---|---|---|
| i. | Resolution | The machine should be able to display single un-insulated tinned copper wire of 40-SWG or 38-AWG. All penetration and resolution condition should be met without pressing any functional key and should be online. |
| ii. | Steel Penetration | 35 mm thickness or more. |
| iii. | Contrast Sensitivity | 4096 gray level stored |
| iv. | Zoom | Continuous Electronic Zoom facility should be available to magnify the chosen area of an image eight times (8X) or more. Image features shall be keyboard controllable. |
| v. | Video resolution | 1280 x 1024/ 24 bits/pixel colour real time processing. |
| vi. | Video Display | Two 22” high- resolution, low radiation, flicker free display Ergonomic, LCD/LED colour monitor or better. |
| vii. | Computer Processor | Intel latest generation processor as indicated in computer specification |

1. The machine should have features of high penetration function, improved resolution function, high density alert, multi energy X-ray imaging facility where materials of different atomic number will be displayed in different colors to distinguish between organic and inorganic materials. With this method to distinguish high density organic materials including explosives. Machine should have variable color or materials stripping to facilitate the operator to monitor images of organic materials for closer scrutiny. All suspicious items (Explosive High density, material narcotics) should be displayed in one mode and that should be on line.
2. Radiation Safety: The machine must comply with requirements of health and safety regulations with regard to mechanical, electrical and radiation hazards. The radiation level should not exceed excepted health standard (0.1mR/Hr. at a distance of 5 cm from external

housing). Along with bid the supplier / manufacturer should furnish relevant certificate from (AERB) Atomic Energy Regulatory Board of India regarding radiation safety. To comply with all applicable health & Radiation Safety regulations for X-Ray system.

3. The company manufacturing the equipment should have ISO certification for manufacturing and serving of X-ray Screening machine.
4. Film safety: Guaranteed safety for high speed films up to IS0 1600 (33 DIN). The machines should be film safe. In other words photographic films must not be damaged due to X-Ray examination.
5. Machine should be properly sealed from all the sides for pest proof. Dust proof cover is to be provided for covering when system is not in use.
6. Facility for variable contrasts must be incorporated to allow enhancement of lighter and darker portion of the image.
7. The machine should be so designed that software enhancement can be easily implemented to take care of new technique in image processing and pattern recognition.
8. Through put should be 200 bags per hour or more. All software features of machine should be online and password protected.
9. Full diagnostic built in test facility. All models should have software controlled diagnosis report facility and system should give printout if printer is connected.
10. All software features of machine should be online and password protected.
11. Machine should be capable for recalling 20 or more previous images.
12. It should be capable for archiving minimum 20,000 images with date & time stamp.
13. Control desk with security housing and locking provision should be available. The operator personal identification number can be entered through the keyboard with generation of log.
14. Facility of image enhancement should be available.
15. All models should have online recording facility it should be possible to take backup of stored images from X-BIS to CD/DVD/USB Drive in Bitmap/JPEG format.
16. Lead impregnated safety screens should be available at either ends of the tunnel. Idle roller provided at either ends of the tunnel to facilitate placing of baggage at input and output.
17. System should work on one software only. All software features should be controlled from Keyboard of machine only. Keyboard function should be user friendly. To enable/ disable the software features system should not be rebooted.
18. If the machine fails to penetrate a particular item then an alarm video and audio both should be generated to notify the operator.
19. The threat image projection (TIP) system software to be incorporated in all X-Ray BIS operation.
20. Copy of all software including X-Ray software with recovery CD must be provided
21. Operational Training - Operating staff has to be provided free training.
22. Operating & service manual shall be provided with each machine.
23. Other Features to include following:

    i) Edge & Variable edge enhancement.
    ii) Inverse Video

iii) Fast initial warm-up
iv) Pseudo color
v) Date & time display.

24. Computer Specifications (minimum or better to be provided).

- 12th Generation Intel® Core™ i3 Processor with 4.0 GHz, 10 MB cache.
- Intel® HD Integrated Graphics.
- RAM 8GB
- Hard Disk Drive: 1 TB or more 7200 rpm SATA HDD or 500 GB SSD
- Mouse: Optical, Keyboard: windows keyboard.
- Bays: 4 no's (2nos 5.25 inches for Optical Media Drives and 2 No's 3.5 inches for Hard Disk Drives).
- Ports: 6 No's USB_3.0 ports (with at least 2 in front), 1 serial port, Parallel port,1PS/2 Key board and 1 PS2 Mouse port, audio ports for microphone and headphone in front.
- Cabinet: Mini tower.
- DMI: DMI 2.0 compliance and support.
- Operating System: Windows / Linux latest version reloaded with Media and Documentation and certification of Authenticity.
- OS Certification: Window 10 / Linux version Certifications.
- Power Management: Screen blanking, hard disk and system idle mode in Power On, Set up password, power supply SMPS surge protected.
- Preloaded Software: Norton, McAfee, E-trust or equivalent Antivirus(latest version) with license.

25. UPS-3KVA online with backup time of 30 minutes.

Each X-Ray system of the tunnel size 600 (+8%) mm(width) x 400 (+8%) mm(height) shall have an external single phase 3KVA On-line UPS to support PC and electronics. It should be of Standard manufacturer make with maintenance free battery to provide power back-up for at least 30 minutes.

NOTE: SPECIFIED ABOVE ARE MINIMUM REQUIREMENTS.

26. Operator Control Console:

i) Operator interface shall provide a simple, flexible and powerful way for operator to control the system and make maximum use of the imaging information.
ii) The operator interface shall combine not more than three button conveyor control with a touch sensitive pad allowing heads-up operation.

27. Entry/ Exit roller Table:

i) Two stainless steel roller tables shall be provided with each system, one at the entry side and other at the exit side.
ii) Each roller table shall be of at least entry 1000mm and exit 1000mm length. The height should be adjustable according to the height of the system conveyor.

28. Combined Test Piece (CTP):

Each system shall have a Combined Test Piece (CTP) for testing the steel penetration, wire resolution, organic and inorganic material discrimination of the machine. Bidder to provide a test certificate for the accuracy of CTP from a Govt. Laboratory. This has to be provided along with every machine.

## 1.6. SOFTWARE REQUIREMENTS:

### 1.6.1. Image Management Software

It shall offer a complete X-Ray image analysis solution that shall be pre-installed in the system's computer.

1.6.1.1. The software shall provide the following:

- Visual alerts to the operator in case of a threat or high density object.
- Operator logs
- Date, Time and Operator ID
- All image analysis features are available on the control panel and with one button for each function with proper labeling .
- Self-diagnostics tool with error message and system configuration tool.

### 1.6.2. Threat Image Projection (TIP) Software Features:

The TIP software shall perform the following applications:

TIP software facility shall be incorporated in the X-ray machines to assist supervisors in testing the operator alertness and training X-ray screeners to improve their ability in identifying specific threat object. The system shall create a threat object and the same should be superimposed on monitor screen while a bag is being screened. To acknowledge that the operator has seen the false object, operator shall press the control panel key that will cause the computer generated threat object to disappear from x - rayed bag image on the VDU screen. Each operator's action shall be recorded in the hard disc of the computer for the auditing purpose by the supervisor or other authorized person.

1.6.2.1. Design of the system:

TIP software should be compatible with other X - ray technologies such as automatic reject unit, dual X-ray screen technologies, automatic threat recognition system etc. All X-ray image functions shall be available at the same time along with the TIP.

1.6.2.2. Image Library:

a) The TIP facility should have an image library containing at least 100 Explosive devices, 100 knives and 100 firearms in various sizes, shapes, locations & orientations. However, the system shall have facility to expand the library to incorporate additional images by user without assistance of the manufacturer.

b) The image library should contain images of threats at different orientations. Both plane and end-on orientation should be used. Images of threats should be assigned different file names and references; however, it shall be possible to cross- reference these as the same threat. All threat image projection images shall be realistic, representative and non-distinguishable from real threat items.

1.6.2.3. Time Interval:

a) Programming facility shall be available to project threat images at different intervals. The time period for threat image as well as image mix in percentage shall be user programmable e.g. software shall select 40 % images of explosive devices, 35 % of fire arms & 25 % knives or random etc.

b) Once the screener has responded to identity the computer generated threat image, it should remain on the screen for a predefined user programmable time for analysis. The image should be highlighted, upon identification, and feedback message shall be visible to the screener.

1.6.2.4. System Administration:

a) The threat image projection facility shall have details of user database such as station name, screener name, organization, user ID number, level of access such as screener, administrator, maintenance & password protection etc.

b) Access to start up menu should be restricted only to the authorized individuals. A log - in procedure by means of “Password” or Security Key”, should be able to restrict access to each of the comment. The log - in procedure should not take longer than 20 seconds. The system should have facility to bypass the TIP

facility, if programmed so by the system administrator. It should be ensured that the TIP software shall not be hindrance to normal functioning of X-ray machines.

c) When the operator logs - in or logs - out, message should be displayed on X-ray screen to confirm that operator has been correctly logged – in or logged - out.

1.6.2.5. Feed Back Report:

a) The threat image projection should be capable of giving feedback "HIT, MISS or FALSE ALARM" message. No message shall be presented if a screener has correctly passed a clear bag.

b) A "HIT" message shall be presented when a screener has correctly identified a threat image projection image. A "MISS" message shall be presented when screener fails to identify the TIP image. A "False Alarm" message shall be given when screener incorrectly indicate TIP image when in fact no threat image projection is present. The feedback should clearly indicate on screen that a TIP object has been correctly identified / TIP object has been missed / no TIP object was present and all such information should be recorded in the database.

c) Different colour coding shall be used for feedback to the screener. It is recommended that colour code red for "MISS", green for "HIT" and yellow for "False Alarm or Interrupt" should be used.

d) The system shall automatically prepare the daily log of events for each shift and for each screener performance. TIP log shall include particulars of location/station, X ray machine, name of screener, time & date of threat image, weather threat image was successfully identified or missed etc.

e) The report on Threat Image Projection system should have date and time as per requirement, screener particulars, and decision/ outcome i.e. MISS, HIT or False Alarm in percentage as well in absolute numbers, number of bags screened, categories such as explosive devices, knife or weapon etc.

f) As a standard practice, daily / weekly / monthly report shall be retrieved. Report shall be for any given time and period as per command.

g) All data should be stored on the system for a minimum of two months after it has been down loaded or as specified by purchaser. Threat image projection data on the X-ray machine shall be read only file so that no individual, regardless of access rights to the threat image projection components shall be able to delete or amend any of threat image projection data or time i.e. Threat Image Projection data on the actual XRay machine will be read only file.

## 1.7. COMBINED TEST PIECE (CTP) TESTS REQUIREMENTS:

### 1.7.1. Single Wire Resolution (Test No. 1):

The requirement is to display 38 AWG guaranteed (40 SWG typical) wire not covered by step wedge. A tick should indicate the visibility of appropriate wire. A set of un-insulated tinned copper wire of different sizes in SWG should be placed on a Perspex sheet. Wires are to be laid out in S shaped curves. The wires should be placed behind varying thickness of aluminum. Metallic marker should be provided using high density material, so that SWG numbers in the Video Display Unit (VDU) are clearly visible.

### 1.7.2. Useful Penetration (Test No. 2):

The test defines what level of details can be seen behind a thickness of known material. The CTP should have different gauges of wire behind varying thickness of aluminum. The requirement of this test is that the 26 SWG wire should be seen under second step wedge (5/16”).

### 1.7.3. Material discrimination (Test No. 3):

The requirement is that different colors should be allocated to the sample of organic and inorganic substances. With multi energy X-Ray, it should be possible to distinguish between materials of different average atomic number for differentiating between organic and inorganic substances. The use of sugar and salt samples encapsulated on the test piece and various materials used in the construction of CTP should check the material discrimination facility.

### 1.7.4. Simple Penetration (Test No. 4):

The requirement is that the lead should be visible beneath 26mm of steel. This test defines what thickness of steel the machine should be able to penetrate.

### 1.7.5. Spatial Resolution (Test No. 5):

The requirement of this test is that vertical and horizontal grating to be seen. This test defines the ability of the system to distinguish and display objects which are close together. The CTP should have 16 copper gratings at right angles to each other.

1.7.6. Thin Metal Imaging (Test No. 6):

The requirement of this test is to image 0.1 mm thick steel. This tests the machine ability to image thin metal. A number of thin metals strips of various thicknesses should be placed in a row.

1.7.7. Method of Testing:

a) The CTP should be used as a quick routine test to be carried out daily to ensure that equipment is working properly and satisfactory image is obtained.
b) The CTP should be placed on the belt and passed through the belt at least once in a day before the baggage is screened or after the X- Ray equipment is switched on to ensure that the equipment is working properly and image is satisfactory.

c) The CTP should be viewed by using image enhancing facility till the operator is satisfied that the machine is working properly. The optimum position of CTP on the belt will depend on X-Ray source and detector arrangements. This should be ascertained from the service engineer, if need arises.

1.7.8. Results:

The best result out of multiple scanning of CTP in both colour and black and white images shall be taken in to account for a particular machine.

## 1.8. TEST REQUIREMENTS:

a) Conditions of Tests:
   i) Unless otherwise specified all tests shall be carried out at ambient atmospheric conditions.
   ii) Inspection and testing shall be carried out to the effect that all requirements of this specification are compiled with.
   iii) Inspection shall be carried out for complete unit of X-Ray Baggage Screening System along with associated accessories.

b) Type Tests:

For type test, one complete system of X-Ray Baggage Screening System shall be subjected to following tests as applicable

a) Visual Inspection.
b) Performance Test.
c) Factory Acceptance Test.

d) Endurance Test.

c) Acceptance Tests:

The following shall constitute the acceptance tests which shall be carried out by the inspecting authority for the purpose of acceptance of complete systems in each site offered by the supplier.

a) Visual Inspection of complete system.
b) Performance Test.
c) Endurance Test.

Any other tests shall be carried out as considered necessary by the inspecting authority

d) Routine Tests:

The following shall comprise the routine tests and shall be conducted by manufacturer on every equipment and the test results will be submitted to the inspection authority before inspection. The application software in proper format, wherever applicable, shall also be submitted to the inspecting authority in advance.

a) Visual inspection of complete system
b) Performance Test

Any other tests shall be carried out as considered necessary by the inspecting authority.

**1.9. TEST PROCEDURE**

The test procedure shall be based on the published system design. The methodologies to be adopted for various tests shall be decided taking into account the system design/ configuration.

Visual Inspection

Each equipment of X-Ray Baggage Screening System shall be visually inspected to ensure compliance with the requirement of clauses mentioned in this specification. The visual inspection shall broadly include:

a) System Level checking:

i) Constructional details
ii) Dimensional check
iii) General Workmanship
iv) Configuration

b) Performance Test:

i) The equipment shall comply with the requirements as specified in clauses of technical specification.

ii) The PC Workstations for use of operators/ screeners and the administrators as mentioned in technical specification shall be of standard make only complying with all the parameters mentioned in the specification.

c) Endurance Test:

During type test, endurance test shall be conducted on complete system for continuous operation which shall be 168 Hrs. at ambient room temperature without giving any deterioration in performance.

During acceptance test, endurance test shall be conducted on complete system for continuous operation which shall be 48 Hrs. at ambient room temperature without giving any deterioration in performance.

## 1.10. QUALITY ASSURANCE:

a) All materials and workmanship shall be of good quality.

• Since the quality of the equipment bears a direct relationship to the manufacturing process and the environment under which it is manufactured, the manufacturer shall ensure Quality Assurance Program of adequate standard.

• Validation and system of monitoring of QA procedure shall form a part of type approval. The necessary plant, machinery and test instruments as mentioned in Technical Requirements shall be available with the manufacturer.

b) Along with the prototype sample for type test, the manufacturer shall submit the Quality Assurance Manual.

c) The availability of machine during DLP & CAMC shall not be less than 99% this will be calculated in following manner:

$$\frac{\text{Total Time - Down Time x 100}}{\text{Total Time}}$$

Total time taken shall be revenue hours of operation of KOCHI Metro. Operation hours are from 05:45 AM to 11:00 PM every day.

## 1.11. TRAINING.

a) On-site training shall be provided to the KMRL staff/ security personnel which shall include complete assembly of each types of system through the use of various modules, integration of hardware with software and complete operation of the system.

b) At the time of handover of the X ray baggage scanning system, concerned station staff in each station needs to be trained for operation and fault diagnosis etc. Apart from this, training for operators of the machine needs to be organised. No additional payment will be made for this training.

c) Sets of training manual in two hard copies and two soft copies for each equipment containing details of the technical specifications, installation and commissioning, troubleshooting & maintenance schedule etc. shall be supplied along with the system.

1.12. Spares and Consumables

Consumables: The Vendor is required to supply all consumables of X-Ray machine (brand new & unused) free of cost during DLP + CAMC period.

Spares: The Vendor is required to supply all spares of X-Ray machine (brand new & unused) free of cost during DLP + CAMC period.

To reduce downtime, the vendor is required to store at his own premises locally in Kochi & supply all spares consumables as & when required by KMRL during DLP + CAMC period.

## 2. MULTI -ZONE DOOR FRAME METAL DETECTOR (DFMD)

2.1. SCOPE

The specification of Door Frame Metal Detector covers technical specification and requirement of screening systems for persons consisting of multi-zone, highly sensitive door frame metal detector and associated accessories at proposed locations.

The door frame metal detector should be capable of helping operators/screeners to easily differentiate between harmless objects and potential threats.

2.2. SYSTEM DESCRIPTION:

Door Frame Metal Detector (DFMD) shall be rugged and sensitive. These walk-through units shall screen people from head- to-toe and shall detect small, hard to- find weapons and threats. It should also detect small stainless steel knife, 9 volt battery, Lithium Ion battery (CR123A) etc. It should be accurate and fast for threat detection and shall not cause hindrance to movement of persons being screened.

The DFMD shall be robust, lightweight, versatile and weatherproof and shall be able to detect even floor level small weapons.

The performance of the DFMD should be independent of the speed of the person passing through.

2.3. GENERAL REQUIREMENTS:

a. The equipment shall adopt widely accepted technology for its operation and shall be of proven performance. The equipment shall conform to international standards/regulations for safety & EMC requirements.

b. The DFMD shall conform to international & national standards for electrical safety.

c. The DFMD should be user friendly & easy to operate. Operation of DFMD shall not be affected by infrared, ultraviolet, electromagnetic or RF radiation. Offered equipment shall comply relevant safety/ immunity standard.

d. The DFMD shall be cardiac pacemaker, magnetic tape & film safe. Manufacturer/ system integrator shall submit certification to this effect with supporting documents to purchaser.

e. The DFMD shall have built-in circuitry for noise suppression.

f. The circuitry of the DFMD shall be easily accessible and programmable.

g. The system should be capable of adapting to diverse operational requirements and evolving security challenges through easy customization of software-based controls and tools.

h. The power supply available shall be 230 V / 50 Hz AC which may vary from 160 Volts to 270 Volts. All modules of the DFMD should work using this power supply only with requisite converters & voltage stabilizers of suitable capacity.

2.4. TECHNICAL REQUIREMENTS:

a. The DFMD shall detect objects of ferrous, non-ferrous, ferrite & alloys. The design of DFMD should be such that its level of performance is constant over long period of time.

b. The DFMD should have multi zone capability with uniform sensitivity in all zones. The DFMD should have minimum 08 (Eight) detection zones. It should have plug –in cards for easy maintenance. The DFMD should be able to provide detection to flat and rod shaped objects, regardless of their orientation.

c. The DFMD should have user-programmable access codes, tamper-proof settings, and multiple channels permitting several detectors to operate simultaneously in close proximity to one another for effective multi-lane performance. It should be automatic and have continuous self-diagnostic testing capability. The settings shall not get lost in event of a power loss or interruption.

d. The system shall have suitable control panel for easy operation and programming. The arrangement shall be such that only authorized person shall have access to such control panel.

e. All the equipments installed outside the control room shall be able to withstand $0^0$C C to $+60^0$ C and humidity of 95% non-condensing.

f. Interference, which is ’mains-borne’ or radiated by an external source, should not cause the DFMD to raise the alarm spuriously. It should be possible to use equipment such as radio, cordless telephone, walkie-talkie sets, X-ray monitors etc. at a distance of one meter from the archway without causing spurious alarms in the DFMD system.

g. Moving metal beyond one meter from the DFMD should not affect performance of the DFMD. It should be possible to move metallic items like trolleys one meter away from the DFMD without the generation of false alarm.

h. There should be both visual and audible alarms. It should be possible to adjust volume of the audible alarm. At its loudest setting, the volume should be adequate to overcome ambient noise present nearby.

i. Detection circuit shall be continuously active for detection of metals and alloys. The equipment shall have comprehensive self diagnostic features that shall be able to pin point the defects by constantly monitoring the internal circuitry, external connections and environment. The DFMD shall also have inbuilt feature of auto calibration.

j. The DFMD should be tamper proof. All settings should be secured with a key lock & two levels of access codes. Further security should be accomplished with a cabinet lock which prevents unauthorized access to physical cables, connectors & electronics.

k. The DFMD should have traffic & alarm counter. The system should also be functional bidirectional. The DFMD shall reset itself within 3 seconds after an alarm condition.

l. The DFMD system should have an in-built memory system of sufficient capacity to maintain data.

m. The DFMD for networked mode shall have control, monitor and analysis interface module to provide remote access to data collection, alarm statistics and real time detector events for monitoring and analysis.

n. The DFMD for networked mode shall be networkable irrespective of the numbers of detectors and should be accessible from a centralized location for monitoring and analysis.

o. The DFMD for networked mode shall enable end users to change the settings from a remote location.

p. The DFMD system for networked mode shall enable end users to integrate it with other security systems like Access Control System, Video Surveillance System etc.

## 2.5. HARDWARE REQUIREMENTS:

Internal dimensions of DFMD shall not be less than 0.72m (breadth) x 2.0m (height) x 0.57m (width).The DFMD should also have in-built rechargeable battery back-up for minimum 4 hours operation in case of interruption of 230 Volts AC power supply.

Alarms & Indications:

There should be both visual and audio alarms. It shall be possible to adjust volume of audible alarm. The setting of the volume shall ensure that alarm is audible despite ambient noise.

Display: A visual display should be located in the overhead panel. The display should provide calibration and operational information, including program and sensitivity settings, operator functions and fault indication. The display should display all self-prompting regulation and control functions as well as traffic count information.

Control Panel: A suitable control panel shall be provided to turn unit on, access and adjust setup and programming.

The control panel shall be used to turn the DFMD on. The unit should be ready to operate within ten seconds. The manual self-test shall be activated at any time by pressing suitable control.
The control panel shall be used to turn the DFMD off, ensuring that all of the information and settings are stored in memory before shut down.

Counter: The DFMD shall have a traffic counter that should track the number of people that have passed through the detector, the number of alarms.. If operator access is enabled, the counter may be resetted by pressing the counter control .

| TECHNICAL SPECIFICATION FOR MULTI-ZONE DOOR FRAME METAL DETECTOR | | |
|---|---|---|
| 1 | Passage Dimension | Height: 200 cm approx<br>Width: 072 cm approx<br>Depth: 057 cm approx |
| 2 | Metal Alarm Indication | a. Visual: Alphanumeric display on the over head control unit. LED zones on left and right side of the DFMD showing height on person bar display (metal locator).<br>b. Audible: Audible tone alert for metal indication. There should be a provision for suitable setting for adjustment of volume of the audible alarm to over corne the ambient noise present in the vicinity. |
| 3 | Zones | Not less than Eight real horizontal detection zones, covering full height of the equipment. |
| 4 | Sensitivity | DFMD should have multi-zone capability with uniform sensitivity in all zones. |
| 5 | Static Metal Compensation | DFMD installed closed to fixed sheet or pieces of metal, which form part of the building or its fittings. The DFMD should compensate for the presence of such metal and its performance should not be degraded by the presence of metal as stated above. |
| 6 | Immunity to external RF interference | i. Should not be affected by external RF transmission and EMI (Supplier shall submit test certificates from NABL (India) / international accredited lab) It should be possible to use equipment such as radio, portable telephone, walkie-talkie sets, X-ray monitors etc. very near to the archway with out causing spurious alarms.<br>ii. Automatic synchronization for DFMDs located close to each other up to a distance of one foot side by side.<br>iii. Moving metal beyond one mtr from DFMD should not affect performance of the DFMD. It should be possible to move metallic |

| | | |
|---|---|---|
| | | items like trolleys, metallic gate opening / closing one mtr away from the DFMD without the generation of false alarm. |
| 7 | Calibration | DFMD shall have inbuilt feature of both manual and automatic calibration. |
| 8 | Traffic Counter | Intelligent traffic counter in alphanumeric display on the overhead control unit. Should allow scanning from both sides with Transceivers/Photocells in the archway. The counter should track the number of people that have passed through the detector, the number of alarms and should calculate the alarm percentage. The counter should be used to obtain an automatic update on the traffic count. |
| 9 | Detection Capability | i. Uniform detection over the entire walk through area from top to bottom<br>ii. Should be capable to detect ferrous and non – ferrous metals including 9mm cartridge and alloy metals concealed in the body of a person when passed through the archway.<br>iii. Should be able to detect multiple metal objects of various weight, size and shape in different zones simultaneously from head to toe.<br>iv. DFMD should reset itself within 3 Sec after alarm condition.<br>v. Unit should have traffic and alarm counter. The equipment should work in bidirectional mode. |
| 10 | Throughput Rate | A minimum of 25 people per minute. |
| 11 | Systems Access protection | Access for systems settings should have password protection. Should have built - in key pad for accessing the system settings. |
| 12 | Health and Safety | i. Conform to relevant electric safety standard (Supplier shall submit test certificates from NABL (India) / international accredited lab)<br>ii. Magnetic field should be harmless to magnetic media, electronic devices and should be film safe, (Supplier shall submit test certificates from NABL (India) / international accredited lab)<br>iii.Magnetic field should be harmless to wearers of pacemaker (heart) and pregnant women. Supporting test certificates from Govt. recognised medical institute /NABL (India) or other international accredited lab as per ICNIRP guidelines.should be attached. |

| | | |
|---|---|---|
| 13 | Accessories to be provided | i. Standard test samples should be provided for testing the DFMDs.<br>ii. Operating manual for user. |
| 14 | Power Supply | 100 - 260 VAC, 50 -60Hz, 12-24 VDC, should be provided with internal battery back up for 4 hours minimum in operational condition. |
| 15 | Operating Temperature | From $0^{o}$C- $60^{0}$C |
| 16 | Weather Proofing | IP 55 or higher |
| 17 | Tamper Proof | The DFMD Control panel should be tamper proof. All settings should be secured with a key lock. Further security should be accomplished with a cabinet lock to prevent unauthorized access to physical cables, connectors & electronics. |
| 18 | Communication | RS 232/Infra-red/ Ethernet |
| 19 | Battery Back up | Should be provided with battery back-up for 4 hours minimum operation in case of power failure. The machine should have low battery indication alarm. |
| 19 | Humidity | Up to 95% Non- Condensing |
| 20 | Weight | less than 100 Kg |
| 21 | Other Features | i. High discrimination between small masses and personal metallic objects.<br>ii. Should not be affected by heavily reinforced floors.<br>iii. Shall be PC compatible<br>iv. No metal shall be able to be tossed without being detected.<br>v. Key pad shall be provided for program settings<br>vi. Performance of the DFMD should be independent of the speed of person passing through. This is particularly important as a person's foot may swing through the archway without touching the ground, or may come to rest on the ground between the archway pillars. |
| 22 | Defect Liability Period & CAMC | Defect Liability Period shall be 2 Years from the date of commissioning. CAMC is for a period of 8 years beyond DLP of 2 years.<br>During the DLP and CAMC all the services, spares, consumables including battery if any shall be supplied by the contractor. |
| 23 | Compliance | NIJ Standard 0601.02 Compliant, CE |

| | | |
|---|---|---|
| 24 | Availability | The availability of machine during DLP & CAMC shall not be less than 99% this will be calculated in following manner:<br>( (Total Time - Down Time) / Total Time ) x 100<br>Total time taken shall be revenue hours of operation of Kochi Metro. Typical hours are from 05:45 AM to 11:00 PM every day. |

The specifications given are minimum and items with higher specification are acceptable at no extra cost.

## B. SCOPE OF WORK

1. The work (on turn-key basis) covered in the tender includes
   a) Supply, Installation, Testing and Commissioning including Training of X-RAY Baggage Scanning system & Door Frame Metal Detector at various metro stations of Kochi Metro Rail Limited at Kochi Kerala, as specified in the purchase order followed by DLP (defect liability period) of 2 Years (24 months) from the date of commissioning.
   b) Comprehensive Annual Maintenance Contract (CAMC includes all consumables, spares , labour cost, any other cost associated with CAMC) for 8 years after expiry of DLP period.
   c) Taking away the old/existing X-RAY Baggage Inspection Scanner system at the metro stations on as is where is basis. The new X-RAY Baggage scanner & DFMD are being purchased as replacement to the existing X-RAY Baggage scanner & DFMD in KMRL metro stations. The contractor has to remove and take away the existing old X-RAY Baggage scanner from the respective metro stations. Proper disposing of the old X-RAY Baggage Inspection Scanners after taking away from KMRL metro station is the responsibility of the contractor.

   Details of work : `

   The tenderer shall execute the work as given below-

| Sl No | Security System | Quantity (Nos) | Work Details |
|---|---|---|---|
| 1 | XRAY Baggage Scanner (XBIS) | 24 | Supply, Installation, Testing and Commissioning including Training at KMRL metro stations as mentioned in the purchase order followed by DLP of 2 Years (24 Months) from the date of commissioning & Comprehensive Annual Maintenance Contract (CAMC) for 8 years (96 months) after expiry of DLP. |
| | | | Taking away and disposal of the old/existing X-RAY Baggage Inspection Scanner systems at metro stations. |

| | | | |
|---|---|---|---|
| 2 | Door Frame Metal Detector (DFMD) | 13 | Supply, Installation, Testing and Commissioning including Training at KMRL metro stations as mentioned in the purchase order followed by DLP of 2 Years (24 Months) from the date of commissioning & Comprehensive Annual Maintenance Contract (CAMC) for 8 years (96 months) after expiry of DLP. |

1.1. All software and firmware upgrades for the system as applicable shall be free of cost during the time of DLP & CAMC. A confirmation letter from the OEM/OEMs shall be submitted.

1.2. The power supply available at the stations shall be 230 V / 50 Hz AC which may vary from 160 Volts to 270 Volts. All modules of the system should work using this power supply only.

1.3. All the modules of X-RAY Baggage Screening System & DFMD shall be modular in construction. In case of up-gradation of such modules in future, it shall be possible to upgrade them without replacing the entire product.

1.4. AC or DC power distribution system shall be provided with MCB protection by the tenderer for the field equipment's and the equipment's in the operation and control center.

1.5. Power/ Data cable laying and routing for security systems is part of the installation. Power/Data cable of appropriate specification is to be laid and to be routed properly using appropriate conduits, cable manager etc from the designated power sockets to the Baggage scanner & DFMD. Industrial grade extension board/power strip shall be provided with the required number of power sockets at the Baggage scanner/DFMD end. All cables including power, data and control cables to be routed properly with appropriate conduits and in a way that the cable route do not cause hindrance to passenger movement.

1.6. All required materials for installation including mounts viz. screws, nuts, clamps, pipes, wires, terminals and miscellaneous materials etc. should be supplied by contractor as directed by KMRL. All the associated cable connectivity towards the functional requirements of the X-Ray baggage scanners & DFMDs in the supply, installation, commissioning and training at various stations are to be borne by the applicant at his own cost.

1.7. The work is to be carried out at the metro stations /places mentioned in the purchase order, in Kochi.

1.8. Refer chapter-3 for specifications of X-Ray baggage scanning system & DFMD.

1.9. The tenderer must give an undertaking that any bugs/defects arising in the software will be attended free of cost and any improvements made to existing features should also be supplied free of cost.

1.10. Only the engineering/technician trained and certified by the OEM/OEMs shall carry out the installation and commissioning of Equipment.

1.11. The tenderer shall also certify that the installation is done in accordance with the installation specified by the OEM.

1.12. Training - On-site training shall be provided to the KMRL staff/ security personnel which shall include complete assembly of each types of system through the use of various modules, integration of hardware with software and complete operation of the system. Apart from this, training for operators of the machine needs to be organised. No additional payment will be made for this training.

1.13. Hard copy & soft copy of Training manual containing details of the technical specifications, installation and commissioning, troubleshooting & maintenance schedule, operation of machine etc. shall be supplied for each XBIS & DFMD machines.

1.14. In addition to training manual, Hand Book (in hard copy) describing the operation of XBIS machine need to be provided with each XBIS machine for the fast & easy reference of person operating the machine.

1.15. The contractor shall remove and take away the old XBIS machines from the metro stations. Loading, unloading, transportation from the site, all expenses involved etc shall be taken care by the contractor. It is the responsibility of the contractor to follow all rules, regulations, statues in transporting, disposing safely the old XBIS machines.

1.16. A display board of suitable size displaying the list of prohibited materials need to be supplied and fixed over the XRAY Baggage machine. Size & content of the display board shall be shared by KMRL.

1.17. The X ray baggage Operator console need to be an independent unit (not integrated with the X Ray Baggage Scanner unit) so that the Baggage scanner can be placed between the Door Frame Metal Detectors. Bidder may please visit the KMRL metro stations to see the present arrangement of Baggage scanners & DFMDs. The same arrangement need to be followed when replacing the Baggage scanners & DFMDs.

1.18. The contractor may advice, guide and assist KMRL in obtaining Regulatory Consents, licenses, approvals, if any, required from Regulatory bodies, any other Government authorities as per the existing rules, regulations in purchasing, installing, operating, disposing off XRAY baggage scanners and DFMDs. During the DLP/CAMC period, contractor shall also give assistance to KMRL in complying with any new rules, modifications/amendments to existing rules of Regulatory bodies and various other applicable Government authorities.

## 2. SCOPE OF DLP & COMPREHENSIVE ANNUAL MAINTENANCE

The tenderer to ensure CAMC for 08 years (Eight years) after DLP period of 02 years (TWO years) including Hardware, software, consumables, spares, batteries etc.

2.1. Preventive Maintenance service ONCE in 3 months (quarter), this will include-

i. Detailed software check
ii. Detailed hardware check.

iii. Analysis of Alarms logs, Event logs and System Performance and taking corrective action.

2.2. Corrective Maintenance of the complete system (XRay Baggage Scanner & DFMD) including all hardware & software of the system. Corrective Maintenance shall be done for all types of faults, problems experienced and as reported by KMRL representative. The applicant shall attend failure/fault calls on immediate basis and there will be no limit for such calls.

For corrective maintenance where hardware part is faulty and need to be replaced, the contractor has to immediately replace faulty part with healthy spare. It is the responsibility of the contractor to dispose off the defective/faulty parts.

2.3. The KMRL shall inform the contractor through telephone or through writing whenever problem occurs in proper functioning of equipment. The contractor shall confirm by issue of suitable case number regarding the complaint. The contractor must give the Telephone number, Email on which the contractor may be contacted during any time of the day. Complaints need to be attended 24X7 hrs. Complaint escalation matrix also may be shared by the contractor.

2.4. The X-RAY Baggage Screening System, Door Frame Metal Detector or its sub-systems shall be attended as early as possible from the instant information is given by KMRL representative by Phone /Email. During CAMC/DLP period, maximum restoration time shall not exceed 24 hours for X-RAY Baggage Screening System, Door Frame Metal Detector or its sub system.

2.5. Complete hardware and software support to be extended by the applicant.

2.6. The contractor shall carry out any software changes/updates/upgrades required during the DLP/CAMC period.

2.7. KMRL shall maintain day to day failures and problems noticed in digital form and shall be informed to the contractor giving date and time. The contractor's Engineer/ representative has to revert with the cause of failure as well as corrective measures taken by the contractor's engineer with date and time .

2.8. For reducing the downtime, trained and supervisory KMRL staff shall be permitted to minor urgent changes for which suitable log will be maintained by KMRL staff.

2.9. All the tools and testing instruments etc. required for checking testing and attending to routine maintenance and breakdowns shall be arranged by the successful bidder.

2.10. The contractor's staff will carry the routine spares required for preventive maintenance to ensure minimum down time without any additional cost. In addition to those spares applicant will also arrange other spares if required without any additional charges.

2.11. In case of major repairs necessitating removal of the equipment to the applicant's service center, the system or its parts to be reinstalled at the KMRL premises after repairing the set in working condition on immediate basis.

2.12. If the repair time is likely to exceed 24 hrs. then the applicant will provide a standby equipment at applicants cost.

2.13. Only trained engineer/technician shall carry out the maintenance of Equipment.The applicant shall depute only competent and efficient staff for routine maintenance as well as the break downs.

2.14. Contractor shall furnish to the KMRL the Name, Addresses and Telephone number of the Engineers responsible for the maintenance work. Name and telephone Number of Engineer who can be contacted after office hours and Sunday/Holidays shall also be furnished to the KMRL. Applicant shall provide identification to their Engineers and staff authorized to attend the maintenance work so as to facilitate verification by representative of KMRL.

2.15. Communication handed over to the applicant's personal at site / office shall deem to have reached their office.

2.16. At the end of each major break down repair, applicant's Engineers should prepare a service report and submit the same to the office of the authorized KMRL person concerned.

2.17. The contractor has to provide all spares and consumables of XRay Baggage Scanner & DFMD free of cost during DLP/CAMC.

2.18. To reduce the downtime of machines during DLP/CAMC, the contractor shall always keep ready minimum quantity of all required spares and consumables of XRay Baggage Scanner & DFMD locally at a ware house in Kochi.

2.19. During the contract period (DLP/CAMC), for the operational convenience, KMRL may need to change the location of DFMD/Baggage scanner (to other stations/ metro locations within Kochi). In such cases shifting shall be done by KMRL. The contractor shall assist KMRL in shifting and in disassembling/reassembling at the locations. The contractor shall continue with the DLP/CAMC maintenance of the machine at the new location.

# CHAPTER -4
# TERMS AND CONDITIONS OF CONTRACT

## 1. DEFINITION AND INTERPRETATION

In the contract, unless the context otherwise requires:

a) "Award of Tender' means the letter or memorandum communicating to the Contractor the acceptance of his tender.

b) "Consignee" means where the equipment are required by the acceptance of the tender to be dispatched by rail, road, air or streamer, the person specified in the Acceptance of tender to whom they are to be delivered at the destination stations.

c) "Contract" means and includes Tender Invitation, Instructions to Tenderers, Tender, Acceptance of Tender, Terms and Conditions of Contract, Scope of Work, Technical Specification and Annexures particulars and the other conditions specified in the acceptance of tender, and amendments.

d) The "Contractor'/Bidder/Tenderer" means the person, firm or company with whom the order of the supply is placed/participated/intend to participate in the tender.

e) "Drawing" means the drawing or drawings specified in or annexed to the specification.

f) The "Inspecting Officer" means the person/team of KMRL specified in the contract for the purpose of inspection of equipment and includes his/their authorized representatives. g) "Particulars" include:-

i. Specifications

ii. "Proprietary mark" or "brand" means the mark and brand of the product which is owned by an industrial firm;

iii. Any other details governing the construction, manufacture or supply of stores as may be prescribed by the contract.

h) "Inspection Test" means such test or tests as are prescribed by the specifications to be made by the Employer or his nominee during installation, and commissioning.

i) "Purchase Officer" means the officer signing the acceptance of tender and includes any officer who has authority to execute the relevant contract on behalf of the Employer;

j) The "Employer" means Kochi Metro Rail Ltd (KMRL).

k) "Signed" includes stamped, except in the case of acceptance of tender or any amendment thereof;

l) "Site" mean the station and Depot buildings location as specified in the work order at which equipment is required to be delivered /installed by the Contractor under the contract or any other place approved by the Employer for the purpose in Kochi

m) "Material" means the goods in the contract, which the Contractor has agreed to supply under the contract;

n) "Test" means such test as is prescribed by the particulars or considered necessary by the KMRL whether performed or made by the Inspecting Officer or any agency acting under the direction of the Inspecting Officer.

o) “Work" means all the work specified or set forth and required in and by the said specifications, drawings and "schedule of Requirements", hereto annexed or to be implied there from or incidental thereto, or to be hereafter specified or required in such explanatory instructions and drawings (being in conformity with the said original specifications, drawings and "Schedule of Requirements").

p) The delivery of the equipment shall be deemed to take place in accordance with the terms of the contract, after approval by the Inspecting Officer /team from KMRL /its representatives the consignee at various sites mentioned in the Scope of work/Purchase Order.

q) “Writing” or “Written" includes matter either in whole or in part, in manuscript typewritten, or printed as the case may be.

r) Terms and expression not herein defined shall have the meanings assigned to them in the Sale of Goods Act, 1930 or the Indian Contract Act, 1872 or the General Clauses act, 1897 as the case may be.

s) "Facilities" means the Equipment to be supplied and installed as well as all the Installation Services to be carried out by the Contractor under the Contract.

## 2. SCOPE OF WORK

Technical specifications and Scope of work is detailed in Chapter 3 of this RFP

## 3. CONTRACT DOCUMENTS

Subject to Article Order of Precedence of the Agreement all documents forming part of the Contract (and all parts thereof) are intended to be correlative, complementary and mutually explanatory. The Contract document shall be read as a whole.

## 4. SIGNING AND ACCEPTANCE OF LETTER OF ACCEPTANCE / WORK ORDER & SIGNING OF AGREEMENT

a) Upon receipt of the GeM contract/Work order the Successful Bidder shall provide order acceptance in GeM within 7 (seven) days from the date of issue of contract/Work order.

b) Tenderer will be required to execute the Contract Agreement. The Employer shall prepare the Agreement in the format included in this Document, duly incorporating all the terms of agreement between the two parties. Within 30 days from the date of issue of the letter of acceptance, the successful tenderer will be required to execute the Contract Agreement. The performance guarantee (Security) should be submitted immediately after issue of letter of acceptance but not later than the agreement is signed between the parties. One copy of the Agreement duly signed by the Employer and the contractor through their authorized signatories will be supplied by the Employer to the Contractor.

Failure of the successful Tenderer to submit performance guarantee within the specified period shall constitute sufficient grounds for the annulment of the award and forfeiture of the Tender Security. Failure of the successful Tenderer in executing the agreement within the specified period shall constitute sufficient grounds for the annulment of the award and forfeiture of the Tender Security.

c) In case the Successful Bidder fails to accept the Work Order/Purchase Order and/or complete the work within the time limit specified in the Work Order/Purchase Order or any extension thereof as may be decided by KMRL, without any additional cost to KMRL, the Bid Security shall be forfeited.

d) The agreement shall be entered in non-judicial stamp paper (duty) of value "One Rupee for every 1000 Rupees or part thereof on the amount agreed in the contract, subject to a minimum of rupees 500 and a maximum of rupees one lakh". The cost of stamp paper is to be fully borne by the successful bidder.

5. **SEVERABILITY**

If any provision or condition of the Contract is prohibited or rendered invalid or unenforceable, such prohibition, invalidity or unenforceability shall not affect the validity or enforceability of any other provisions and conditions of the Contract.

6. **CONTRACTOR'S RESPONSIBILITIES**

a) The Contractor confirms that it has entered into this Contract on the basis of a proper examination of the data relating to the Facilities provided by the Employer, The Contractor acknowledges that any failure, to acquaint itself with all such data and information shall not relieve its responsibility for properly estimating the difficulty or cost of successfully performing the Facilities.

b) The contractor should use materials and workmanship as per the industry standards and best practices and keeping the aesthetic look of the premises.

c) The Contractor shall comply with all laws, which are in force in India. The laws will include all local, State, National or other laws that affect the performance of the Contract and bind upon the Contractor. The Contractor shall indemnify and hold harmless the Employer from and against any and all liabilities, damages, claims, fines, penalties and expenses of whatever nature arising or resulting from the violation of such laws by the Contractor or its personnel, including the Subcontractors and their personnel.

d) All costs, damages and expenses which Kochi Metro Rail Limited may have incurred or suffered and which are recoverable from the contractor under the terms of this contract or the relevant law may, at the discretion of the Kochi Metro Rail Limited be recovered by deducting the requisite amounts from any moneys due and payable or refundable to the contractor on any account whatsoever or by legal proceedings. The Kochi Metro Rail Limited also reserves the right and shall be entitled to retain payments due to the contractor under this contract and to set off the same against all claims whether arising out of this contract or out of any other transaction whatsoever against the contractor.

e) The Contractor/Bidder shall design, manufacture, deliver, supply, install, commission XBIS machines at the locations/metro station specified by KMRL and shall provide training.
f) The Contractor/Bidder shall remove and take back the existing old XBIS machines from the locations/metro stations wherever old machines are being replaced with new machines as per the contract.
g) The Contractor/Bidder shall carry out Defect Liability Period (DLP) and Comprehensive Annual Maintenance contract (CAMC) obligation (including associated purchases) with due care and diligence in accordance with the Contract.

**7. CONFIDENTIAL INFORMATION**

a) The Employer and the Contractor shall keep confidential and shall not, without the written consent of the other party hereto, divulge to any third party, any documents, data or other information furnished directly or indirectly by the other party hereto in connection with the Contract, whether such information has been furnished prior to, during or following termination of the Contract.
b) The Contractor shall not use such documents, data and other information received from the Employer for any purpose other than the design, procurement of Material, construction or such other work and services as are required for the performance of the Contract.

8. **MATERIALS & SERVICES**

a) Materials to be supplied/work to be carried out are indicated in scope of work. The contractor shall take delivery of the materials to locations as mentioned in the scope of work/WO/LoA.
b) All the Materials to be supplied/used for execution of work shall be as per technical specification. The contractor has to load, unload and transport the same to the worksite, installation location at his own cost. No additional amount will be paid by KMRL. Further, care needs to be taken while shifting of the material to the location so that other installations at the stations, viaduct or any other assets of KMRL are not damaged. If any damage happens, the contractor has to make good of the same or the amount will be deducted from the invoices.
c) The contractor will, however, have to procure all the materials required for executing the labour portion of the work/supply of material. The contractor shall engage his own labour and supervisor for the execution of the work covered in the contract.
d) The contractor shall supply all the materials as mentioned in schedule and transport the material to the location at his own cost. However, any other material not specifically mentioned herewith but is required for executing the scope of work covered in the tender shall be provided by the bidder at his own cost.
e) All required materials if any for the technical specification mentioned in Chapter 3, Technical specification including mounts viz. screws, nuts, clamps, pipes, wires, terminals and miscellaneous materials etc. should be supplied by contractor as directed by KMRL.

f) All the materials and equipment to be supplied and used for execution of work shall be of Industrial grade. The tenderer has to load/unload and transport/move the equipment to the worksite and to the designated location in station at his own cost. No additional amount will be paid by KMRL.

9. **CONTRACT**

a) This contract is for the scope of work mentioned in Chapter-3. All Material must be brand new and unused. Unpacking/seal opening has to be done in presence of KMRL/authorized representatives.

b) The whole contract is to be executed in the most approved, substantial and workmanship manner, to the entire satisfaction of the Employer or his nominee, who, both personally and may his deputies, shall have full power, at every stage of progress, to inspect the material at such times as he may deem fit and to reject any of the material which he may disapprove.

10. **KMRL's RESPONSIBILITY**

a) Access to the respective Metro stations will be provided by KMRL.

b) In case of any Force Majeure event, the Employer shall promptly notify the Responsible Engineer and the Contractor of any proposals for overcoming the consequences of the Force Majeure.

11. **EMD**

Refer clause 7 of Chapter 2

12. **Release of EMD**

Refer clause 7 of Chapter 2

13. **TIME AND SCHEDULE FOR DELIVERY**

DELIVERY - All equipment in full needs to be delivered at the respective Kochi Metro stations as per the Contract before 28/03/2025.

The location/stations and location wise quantity is given below-

| SL No | LOCATION / STATIONS | XRAY Baggage Scanner | | DFMD |
|---|---|---|---|---|
| | | Quantity of new XBIS machines to supply, install, test & commission at the location (Nos) | Quantity of old XBIS machines to take away from the location (Nos) | Quantity of new DFMD to supply, install, test & commission at the location (Nos) |
| 1 | ALUVA | 2 | 2 | 0 |

| | | | | |
|---|---|---|---|---|
| 2 | PULINCHODU | 1 | 1 | 0 |
| 3 | COMPANYPADY | 1 | 1 | 0 |
| 4 | AMBATTUKAVU | 1 | 1 | 0 |
| 5 | MUTTOM | 1 | 1 | 0 |
| 6 | KALAMASSERY | 1 | 1 | 0 |
| 7 | CUSAT | 1 | 1 | 0 |
| 8 | PATHADIPALAM | 1 | 1 | 0 |
| 9 | EDAPALLY | 2 | 2 | 0 |
| 10 | CHANGAMPUZA PARK | 2 | 2 | 0 |
| 11 | PALARIVATTOM | 2 | 2 | 0 |
| 12 | JLN STADIUM | 2 | 2 | 2 |
| 13 | KALOOR | 1 | 1 | 2 |
| 14 | TOWN HALL | 2 | 2 | 4 |
| 15 | MG ROAD | 2 | 2 | 2 |
| 16 | MAHARAJAS COLLEGE | 2 | 2 | 3 |

KMRL may change the locations, extend the delivery date or may ask for staggered delivery for the convenience of installation, replacement works in stations. In case of extension of delivery time, staggered delivery, KMRL shall issue request in writing and the contractor has to follow the delivery locations & dates.

The delivery, installation and commissioning need to be done in KMRL metro stations which are operational. Hence unloading, shifting inside metro station building, installation, commissioning need to be done during non-revenue hours (during night 11PM to 5AM) of KMRL metro operations. The contractor need to plan and complete the work accordingly and without affecting the metro operations, passenger movement in stations, and without damaging any KMRL property. Damages/losses to KMRL, if any, shall be recovered from the contractor from the payments due/invoices.

TIME FOR INSTALLATION AND COMMISSIONING - The installation, commissioning & training of new XRAY Baggage Scanners & DFMDs needs to be completed before 28/04/2025.

The old XBIS machine needs to be removed from the metro station within 1 week after the date of installation of the new XBIS machine.

Care needs to be taken while shifting the equipment to/from the stations so that other installations at the stations, flooring of the stations are not damaged. If any damage to other installations, station building happens, the applicant has to make good of the same or the amount will be deducted from the invoices.

14. **DELIVERY / COMPLETION**

a) Notification of delivery and dispatch with regard to each and every consignment shall be made to the Purchaser immediately on dispatch and delivery. The tracking number of transport mode along with necessary details for tracking of the dispatched material needs to be communicated to the purchaser on every dispatch.

b) Time for delivery/completion: Time is the essence of the contract. The time and date specified in the contract or as extended for the delivery of the Material/completion of work shall be deemed to be the essence of the contract and delivery must be completed not later than the dates so specified or extended by Purchaser. Unless the delivery/completion period is extended by KMRL in writing, the contract/PO is deemed to be cancelled and KMRL may forfeit Performance security deposit fully.

c) Progress of Deliveries: The contractor shall allow reasonable facilities and free access to his works and records to the inspecting officer, progress officer or such other officer as may be nominated by the Purchaser for the purpose of ascertaining the progress of the deliveries under the contract.

d) Extension of time for Delivery/Completion: The purchaser may extend the timeline for delivery of Material/completion of work at his own discretion due to the reasons which may be beyond the control of purchaser. The price quoted shall remain same even in case of extended delivery time line. The purchase order will remain null and void after the delivery date mentioned in the PO/WO/LOA. Delivery against the invalid Purchase order/work order will not be accepted unless the supplier requested to extend delivery date and the same is approved by KMRL in writing. The Liquidated Damages clause will anyways will be applicable to the initial delivery date before extension. However in case of extended delivery, the DLP and CAMC period calculation if any as per the scope of this tender will be done as per the acceptance certificate issued by the purchaser.

e) The Contractor shall be required by the Purchaser to deliver the equipment on Delivery Duty Paid (DDP) basis at KMRL Metro stations, the quantities of the equipment detailed therein shall be delivered not later than the dates specified in the purchase order/LoA. The delivery will not be deemed to be complete until and unless the equipment are inspected and accepted by the Inspecting Officer/team of KMRL or its representatives.

## 15. FAILURE AND TERMINATION

If the seller does not perform its obligations within the Delivery Period / Date mentioned in the Contract, the same would constitute the breach of the Contract and the Buyer shall have the right to Cancel or Withdraw the Contract for the unsupplied portion after the expiry of the original, extended or re-fixed delivery date or period stipulated in the Contract. Such cancellation of contract on account of non - performance by the Seller would entitle the Buyer to forfeit the performance security besides other actions such as downgrading the Seller's rating or debarment from the GeM for specified period as decided by GeM on merits.

## 16. CONSEQUENCES OF REJECTION

a. If on inspection of the material at site, are found to be not matching the requirement of purchaser as mentioned in the tender document and are being rejected by the Inspecting Officer/team of KMRL or its representatives, the Contractor would be required to make

satisfactory supplies of brand new and unused Material meeting the requirement as mentioned in the tender document within the stipulated period of delivery.

b. Removal of rejected stores:

i. On rejection of any Material submitted for inspection at a place other than the premises of the Contractor, such stores shall be removed by the Contractor at his own cost subject as herein after stipulated, within 10 days of the date of intimation in writing of such rejection.

ii. All rejected Material shall in any event and circumstances remain and always are at the risk of the Contractor immediately on such rejection. If such Material are not removed by the Contractor within the periods aforementioned, the Inspecting Officer/KMRL representatives or its authorized personnel may remove the rejected Material. The purchaser shall, in addition, be entitled to recover from the Contractor the handling and storage charges on the rejected Material after the expiry of the time-limit mentioned above.

17. **FORCE MAJEURE**

In the event of any unforeseen event during the currency of the Contract, such as earthquake, war, fires, floods, or acts of God, as a result of which, either party (Employer/contractor) is prevented, or hindered in performing any of its obligations under the contract, then it shall within a week from the commencement thereof, notify the same in writing to the other party with reasonable evidence thereof. If the force majeure condition(s) mentioned above be in force for a period of 90 days or more at any time, the either party shall have the option to terminate the contract on expiry of 90 days of commencement of such force majeure by giving 14 days' notice to the other party in writing. In case of such termination, no damages shall be claimed by either party against the other, save and except those, which had occurred under any other clause of this contract prior to such termination.

18. **INVOICE/BILLS**

Invoices submitted by successful bidders should be E invoices (i.e, Invoices electronically uploaded and authenticated with a unique IRN and digitally signed QR code). In case E invoicing is not applicable, e–Invoice exemption declaration form shall be submitted by vendor along with invoice. If the above mentioned form cannot be filed due to non-enablement of e- invoicing as turnover is less than 5 Cr., proof of the same shall be submitted along with invoice.

19. **ACCEPTANCE TESTS**

1. Immediately after the completion of the work at each station the applicant shall certify and advise the purchaser in writing that the installation is:

a. Complete
b. Ready for satisfactory commercial service and
c. Ready to be handed over.

2. Any component, modules, sub-assemblies or equipment failing during the commissioning test shall be replaced/repaired free of cost by tenderer.
3. KMRL shall issue an Acceptance certificate for successful commissioning of the section covering all the materials and services included in the scope of works after the final acceptance test as per the approved test procedure have been completed and the performance has been found to meet the specifications. KMRL decision in this respect shall be final. The Acceptance Certificate shall be signed by both the parties. The period of maintenance (DLP) of works shall commence from the date of issue of Acceptance Certificate.
4. After the work has been completed & placed in service and Acceptance certificate issued by KMRL, the applicant shall be responsible for proper maintenance supervision of the work for a period of twenty four months from the date of commissioning (DLP period). For this purpose he shall prepare a maintenance plan and make available the services of qualified maintenance engineer stationed at the location approved by KMRL who will guide and supervise the work of maintenance staff. The maintenance engineer of the tenderer will visit the total installation at least once in a month
5. During this period of maintenance supervision if any lacuna is noticed in the functioning, as a result of any work, the applicant will rectify the same free of cost. During such rectification if any faulty equipment/modules need replacement or repair, they shall be provided by the applicant from the set of equipment or modules that the applicant should bring to the site of installation in addition to all the materials to be supplied against this contract.
6. After the DLP period, the applicant has to maintain the system under CAMC for 8 years.

## 20. DOCUMENTS TO BE FURNISHED ALONG WITH EQUIPMENT AT THE TIME OF INSTALLATION

The following technical documents and other information to be furnished along with the equipment are required to be submitted in the technical bid

1. Documents for installation details etc.

2. Manual for operation and maintenance of the equipment supplied.

3. Details of test accessories and test facilities provided along with write up on testing and troubleshooting procedures. Technical literature of equipment to be supplied, along with the supply of materials.

4. All necessary software/ Originals CD/DVDs which are part of technical equipment.

5. Guarantee/Warranty certificate to include the Two year DLP period

6. List of consumables and spare parts per equipment.

## 21. PERFORMANCE BANK GUARANTEE AND RETENSION

### E. Performance Bank Guarantee

The successful bidder shall furnish to KMRL a security deposit in the form of Payment online through RTGS/ internet banking or PBG for an amount, equivalent to 5% of the Contract Price within 21 days from the date of issue of GeM contract. The validity of the Bank Guarantee shall be two months (120+2=122 months) beyond the expiry of contract period. Online payment shall be in Beneficiary name KOCHI METRO RAIL LTD Account No. 0803201002929, IFSC Code CNRB0000803, Bank Name CANARA BANK, Branch address ERNAKULAM BROADWAY BRANCH.

Successful Bidder must indicate Contract number and name of Seller entity in the transaction details field at the time of on-line transfer. Bidder has to submit proof of the Online Payment Transfer in place of PBG within 21 days of award of contract.

In case if Performance Bank Guarantee (PBG) is submitting, it must be from a scheduled commercial bank in India, payable at a designated bank branch located in Kochi for an amount equivalent to 5% of the total contract value. Further as and when contract value increases, even as a result of revision in applicable taxes, the contractor shall be required to deposit additional BG(s) to ensure availability of Bank Guarantee at the rate of 5 % of contract value with KMRL at all times.

Irrevocable bank guarantee in the prescribed format, given in Appendix 6, issued by a Scheduled Commercial Bank based in India payable at Kochi. The bank guarantee must be issued on the Structured Financial Messaging System (SFMS) platform. The details of Employer's bank are as under:

| | |
|---|---|
| Bank Name | CANARA BANK |
| Branch | BROADWAY |
| IFSC Code | CNRB0000803 |
| A/c. No. | 0803201002929 |
| Bank Address | BROADWAY BRANCH, ERNAKULAM |
| PINCODE | 682031 |

The bank guarantee issued on the SFMS platform shall only be acceptable to the Employer

The Purchaser shall be entitled on his part to forfeit the amount of the Performance Bank Guarantee/ Security deposit in whole or in part in the event of any default, failure or neglect on the part of the Contract in the fulfilment or performance in all respects of the contract under reference or any other contract with the Purchaser shall also be entitled to deduct from the amount of the Performance Bank Guarantee/Security deposit any loss or damage which the Purchaser may suffer.

KMRL at its sole discretion may request the supplier to extend the Performance bank guarantee on the extension of delivery date equal for a period of extended time of delivery. Failing to extend Performance bank guarantee, KMRL shall be entitled to forfeit the performance bank guarantee

PBG/ security deposit will not bear any interest.

PBG/ security deposit will be released to the contractor within 60 days, after the completion of contract period (completion of 8th year of CAMC) and after all contract obligations if any based on the written request from the successful bidder/contractor/vendor. The bidder has to ensure that any penalty amount due to KMRL is paid within one month after the completion of the 8th year CAMC, failing which the PBG will be encashed and the amount, if any, after deducting the penalty will be paid to the contractor

**Note: Separate ePBG is not required for the CAMC. "PBG clauses stated in Additional clause for CAMC shall not be considered."**

**F. Retention:**

a) A retention amounting to 10% of the gross amount of each bill will be deducted at the time of certifying interim payment, for the due execution and completion of the supply. This retention amount is liable to be forfeited, partly or wholly, if the contractor fails to carry out the assignment or to keep up the desired rate of progress as per contract.

b) 10% retention amount will be released as below

i. 5% will be released at the end of first year of DLP period after deducting, if any, charges for delay in attending fault, as penalty and

ii. 5% will be released at the end of second year of DLP and Final Acceptance Certificate after deducting any charges for delay in attending fault, as penalty.

## 22. CONFLICT OF INTEREST

The remuneration of the Tenderer shall constitute the Tenderer’s sole remuneration in connection with this Contract or the Services and, the Tenderers shall not accept for their own benefit any trade commission, discount or similar payment in connection with activities

pursuant to this Contract or to the Services or in the discharge of their obligations hereunder, and the Tenderers shall use their best efforts to ensure that any Personnel either of them, similarly shall not receive any such additional remuneration Neither the Tenderer nor the Personnel of either of them shall engage, either directly or indirectly, in any of the following activities:

(a) During the term of this Contract, any business or professional activities in India which would conflict with the activities assigned to them under this Contract.

(b) The tenderer shall not be one of the following:

(i) A firm which has been engaged by the Employer to provide consulting services for the preparation related to procurement for or implementation of this project.

(ii) Any association/affiliation (inclusive of parent firms) of a firm or an organization mentioned in para (i) above.

(iii) A Tenderer who lends, or temporarily seconds its personnel to firms or organizations which were engaged in consulting services for the preparation related to procurement for or implementation of the project, if the personnel would be involved in any capacity on the same project.

## 23. LATE DELIVERY PENALTY AND LIQUIDATED DAMAGES

a. For Delivery/Work Completion

If the contractor fails to supply the items/complete the work within the stipulated time in the purchase order/work order/LoA and/or contract, the contractor shall pay KMRL in terms of late delivery fee 0.5% of value of delayed goods/work per week of delay subject to maximum 10% of the value of such goods/work. For avoidance of doubt, the L.D will be calculated on pro rata basis for the no. of days of delay for the delayed goods. Any delay happened from KMRL will be excluded for the calculation of L.D

b. Liquidated damages during DLP period & CAMC period

Fault in machine shall be attended within two hours of reporting. The rectification time shall never exceed 24 hours. In case of delay of more than 24 hours, 0.1% of the value of the machine will be levied for every 24 hours of delay in rectification of machine during DLP & CAMC period. The total value of penalty during the contract period will be limited to 10% of contract value. Down time shall be reckoned from the time of intimation of failure to the firm.

## 24. PACKING AND MARKING

Packing: The Contractor shall pack at his own cost the goods sufficiently and properly for transit by rail/road, air and/or sea so as to ensure their being free from loss or damage on arrival at their destination stations as specified in the purchase order. He shall decide the packing for the stores by taking into account the fact that the stores will have to undergo arduous transportation before reaching the destination and will have to be stored and handled in tropical climatic conditions (Including Monsoons) before being put to actual use. Unless otherwise provided in the contract, all containers (including packing cases, boxes, tins, drums and wrappings) in which the stores are supplied by the Contractor shall be considered as non-returnable and their cost as having been included in the contract price. Each packages shall contain a packing note specifying the name and address of the Contractor, the number and date of the acceptance of tender and the Designation of the Purchase Officer issuing the supply orders, the description of the equipment and the quantity contained therein.

Marking : The marking of all goods supplied shall comply with the requirement of the Indian Acts relating to merchandise marks or any amendment thereof and the rules made there under. The following marking of the material is required: - The following particulars should be stenciled with indelible paint on all the materials/packages:

a. Contract No.
b. Station Name
c. Purchaser Name& logo.
d. In addition to the marking as specified above, distinguish color marks should be given so as to distinguish the ultimate Consignees in India.

25. **WARRANTY/DEFECT LIABILITY PERIOD (DLP)**

The period of maintenance (DLP) of works shall commence from the date of issue of Acceptance Certificate. The contractor shall warranty that everything to be furnished under this contract shall be free from defects and faults in design, materials, workmanship and manufacture and shall be of the highest grade and consistent with the established and generally accepted standards for goods of the type ordered and in full conformity with the contract specifications and samples, if any and shall if operable, operate properly. The contractor's liability in respect of any complaints/defect and/or claims shall be limited to the furnished and installation of replacement parts free of any charge or the repair of defective parts only to the extent that such replacement or repairs are attributable to or arise from faulty workmanship of materials or design in the manufacture of the equipment, or wear and tear during normal use, provided that the defects are brought to the notice of the contractor within the warranty period. All replacement and repairs at the KMRL shall call upon the contractor to deliver or perform under this warranty shall be delivered and performed by the contractor promptly and satisfactorily. If the contractor desires, the replaced parts can be taken over by them for disposal as they deem fit within period of one month from the date of replacement. At the expiry of this period, no claim whatsoever shall be on the KMRL. During warranty/DLP the defective parts should be replaced at the earliest in the interest of the contractor to avoid penalty for down time in case of system fails. If the repair is expected to take long time, the contractor must provide a temporary spare from his stock, till they are able to return the original after repairs.

26. **MAINTENANCE DURING DLP & CAMC**

a) The contractor shall visit the site periodically and maintain the systems, subsystems and associated equipment with their maintenance engineer for a period of 2 Years (24 Months) of DLP period from the date of issue of acceptance certificate for the system/sub-system. DLP shall be followed by 8 years of CAMC. The services of maintenance engineer shall be available for the period of DLP & CAMC as and when required by the KMRL representative. The applicant shall ensure that the KMRL personnel are fully acquainted, trained and familiar with detailed procedure to be followed for proper testing, troubleshooting and satisfactory maintenance of the systems and subsystems. In this period of maintenance any defect noticed in the design /manufacturing of the systems shall be attended to promptly and take necessary remedial measure, to replace the defective part free of cost and rectify the system design/manufacturing defect. During this period the applicant should deploy the skilled staff for maintaining all the equipment in good working conditions like cleaning of the system, checking log entry in the server etc. DLP & CAMC works include repair and replacement of defective hardware and software, any future software or firmware up gradation equipment, systems and sub-systems and without any cost of spares.

b) The contractor has to maintain minimum 10% of the spare parts of the equipment supplied to KMRL as on-site spares in Kochi for immediate replacement during failures at the applicants cost to reduce the downtime. The defective parts should be replaced at the earliest in the interest of the applicant to avoid penalty for down time in case of

system fails. If the repair is expected to take long time, the applicant must provide a temporary spare from his stock, till they are able to return the original after repairs.

27. **PAYMENT TERMS**

The standard payment terms subject to recoveries by way of statutory deductions and Liquidated Damages as applicable will be as under: -

a) 100% payment of the supply value of equipment will be made after the successful completion of supply, installation, commissioning and training of X ray baggage scanners & DFMD on submission of following documents after deducting retention amount:

**Documents required**

- Certificate from the purchaser of having receipt of Performance Bank Guarantee.
- Invoice in duplicate.
- Certificate from purchaser/ consignee indicating successful Installation, testing, commissioning of equipment along with necessary operational training to its staff at the site as indicated in purchase order.
- Certificate from purchaser/ consignee that the old equipment is removed from the site.
- Insurance Copy (transit plus storage).
- Packing list.

Note: Retention amount will be deducted as per the tender conditions.

b) Payment for one fourth of the Comprehensive Annual Maintenance Contract (CAMC) value for the respective year including GST shall be made after successful completion of the work for each quarter duly certified by the authorized representative of the KMRL. Payment shall be made on presentation of the following documents:

i. Original Tax Invoice showing the rate and the period for which CAMC charges are claimed.
ii. Preventive maintenance report with filled checklist for the billing period duly signed by KMRL representative.
iii. Certificate of satisfactory Maintenance for the billing period to be given by the authorized KMRL representative
iv. Any other document if requested by KMRL.

Payment of CAMC will be released after all statutory deductions, deduction due to penalty/liquidated damages as applicable. Failure wise total penalty amount for the period will be deducted from the invoice amount.

28. **CONSIGNEE'S RIGHT OF REJECTION**

Notwithstanding any approval which the Inspecting Officer may have given in respect of the stores or any goods or other particulars or the work or workmanship involved in the

performance of the contract (whether with or without any test carried out by the Contractor or the Inspecting Officer or under the direction of the Inspecting Officer) and notwithstanding delivery of the stores /completion/ installation of work where so provided to the interim consignee, it shall be lawful for the consignee, on behalf of the Employer, to reject the stores/work or any part, portion of consignment thereof within 45 days after actual delivery/completion/ installation thereof to him at the place or destination specified in the contract if such stores/work or part, portion of consignment thereof is not in all respects in conformity with the terms and conditions of the contract whether on account of any loss, deterioration or damage before dispatch or delivery or during transit or otherwise howsoever.

29. **RESPONSIBILITY FOR COMPLETENESS**

a) Installation must meet the specifications
b) All the charges incurred towards man-powers, materials, equipments, consumables transportation, movement/shifting within site, disposal of old materials/equipments and making the work useful at the place of installation shall be borne by the successful bidder.
c) Any fittings or accessories which may not be specifically mentioned in the specifications but which are useful or necessary are to be provided by the Contractor without extra charge, and the equipment must meet the operational requirement at the place of delivery.
d) The work (on turn-key basis) shall be performed at the stations specified in the purchase order

30. **INDEMNITY**

a) The prices stated are to include all rights (if any) of patent, registered design or trade mark and the Bidder shall at all times indemnify the Employer against all claims which may be made in respect of the material for infringement of any right protected by patent, registration of designs or trade mark; provided always that in the event of any claim in respect of alleged breach of a patent, registered designs or trade mark being made against Employer, the Employer shall notify the bidder of the same and the bidder shall, at his own expense, either settle any such dispute or conduct any litigation that may arise there from.
b) The bidder shall return all such property and shall be responsible for the full value thereof to be accessed by the Employer whose decision shall be final and binding on the bidder. The bidder shall be liable for loss or damage to such property from whatever cause happening while such property is in the possession of or under the control of the bidder, his servants, workmen or agents.

**31. INSOLVENCY AND BREACH OF CONTRACT**

The Employer may at any time, issue notice in writing summarily terminate the contract without compensation to the Contractor in any of the following events, that is to say:

a) If the Contractor being an individual or a firm: -. Any partner thereof, shall at any time, be adjudged insolvent or shall have a receiving order or order for administration of his

estate made against him or shall take any proceeding for composition under any Insolvency Act for the time being in force or make any conveyance or assignment of his effects or enter into any assignment or composition with his creditors or suspend payment or if the firm be dissolved under the Partnership Act, or

b) If the Contractor being a company is wound up voluntarily or by the order of a Court or a Receiver, Liquidator, or Manager on behalf of the debenture holders is appointed, or circumstances shall have arisen which entitle the Court or Debenture holders to appoint a Receiver, Liquidator or Manager, or
c) If the contractor commits any breach of the contract not herein specifically provided for.
d) Provided always that such determination shall not prejudice any right of action or remedy which shall have accrued or shall accrue thereafter to the Employer and provided also the Contractor shall be liable to pay to the Employer any extra expenditure he is thereby put to and the Contractor shall, under no circumstances, be entitled to any gain on re purchase.

**32. LAWS GOVERNING THE CONTRACT**

a) This contract shall be governed by the Laws of India for the time being in force.
b) Irrespective of the place of delivery and the place of payment under the contract, the contract shall be deemed to have been made at Ernakulam from where the Award of tender has been issued.

**33. ARBITRATION**

In case of any dispute or differences arising out of this agreement then the same shall be settled amicably between the parties by way of mutual discussion or negotiation. If such discussion or negotiation, could not resolve the issue within 30 days from the date of commencement of such discussion or negotiation then the same shall be referred to a Sole Arbitrator to be appointed from the Panel of Arbitrators maintained by KMRL. If any dispute is raised by the Service Provider and arbitration clause is invoked, the KMRL shall within 30 days of such invocation, submit to the Service Provider a panel having not less than 4 Arbitrators, from which the Service Provider shall chose one as Sole Arbitrator. Arbitration shall be conducted under the provisions of the Arbitration and conciliation Act, 1996. Venue and seat of the arbitration shall be Ernakulam. Language of the proceedings shall be English. Award of the sole arbitration shall be final and binding upon the parties.

34. **CORRUPT PRACTICES**

The Bidder shall not offer or give or agree to give to any person in the employment of the Employer or working under the orders of the Employer any gift or consideration of any kind as an inducement or reward of doing or forbearing to do or having done or forborne to do any act in relation to the obtaining or execution of the contract or any other contract with the Employer or Government for showing any favor or for bearing to show disfavor to any person in relation to the contract or to any other contract with the Employer or Government. Any breach of the aforesaid condition by the Contractor, or any one employed by him or acting on his behalf, under Chapter IX of the Indian Penal

Code,1860 or the Prevention of Corruption Act, 1988 or any other act enacted for the prevention of corruption by public servants shall entitle the Employer to cancel the contract and all or any other contracts with the Bidder and to recover from the bidder the amount of any loss arising from such cancellation.If it is found that Bidder(s) had engaged in corrupt/ fraudulent practice in securing and executing the Contracts, KMRL reserves the right:

i. not to award Contracts to such Bidder,

ii. to cancel the Contracts, if already awarded. In case of cancellation, KMRL shall be entitled to recover from the Bidder the amount of any loss arising from such cancellation in accordance with the provisions of the Tender Document. KMRL shall also have the right to forfeit the Bid Security/ Performance Security of such Bidder, and

iii. to ban the business dealing with the Bidder who engaged in such practices either indefinitely or for a specified period of time.

35. **NON EXCLUSIVE**

KMRL on its sole discretion reserves the rights to avail the similar services from other Service Providers/ others during the Contract period.

36. **VARIATION/REPEAT ORDER**

a. VARIATION:

The Employer reserves the right to increase or decrease the quantity up to 25% of the quantity offered by the successful tenderer. The successful tenderer is bound to accept the increase or decrease in the tendered quantity upto 25% under this clause as ordered at the time of placement of contract and/or during the currency of the contract. Employer also reserves the right to operate the increase in quantity upto 25% at its sole discretion within a period of one year from the date of first Purchase Order. While operating this clause the quantity shall be rounded off to the next whole number. PBG may be confiscated by KMRL if contractor/supplier fails to execute according this variation clause.

b. **REPEAT ORDER :**

The Contract will have Repeat Order Clause, wherein KMRL can order under the Contract within six months from the date of completion of supply under the original PO. The Repeat Order will have rates on not exceeding basis while the terms and conditions will remain unchanged. It will be entirely the discretion of KMRL to exercise the Repeat order or not.

37. **EXTENSION OF CONTRACT: -** The CAMC will be for a period of 8 years from the date of completion of DLP. Thereafter on reviewing the satisfactory performance, the CAMC may be extended to further 1 year, based on the recommendation of evaluation committee nominated by KMRL on mutual agreement with rates not exceeding basis while the terms and conditions will remain unchanged. It will be entirely the discretion of KMRL to exercise the extension of contract or not. Validity of PBG must be extended accordingly.

38. **INTEGRITY PACT**

All the Users in GeM i.e. Seller as well as Buyer agree not to indulge in any corrupt practices including without limitation any activity or action to influence the transaction on any aspect of contract and commit to take all measures necessary to prevent corruption maintaining complete transparency and fairness in all activities related to GeM. Users agree to follow and adhere with the Integrity Pact guidelines provided on GeM Portal. Format of integrity pact is enclosed in appendix 13.

# APPENDIX 1

## TENDER INDEX

The following tender index/check list is intended to help the tenderers in submitting offer, which are complete. An incomplete offer is liable to be rejected. Tenderers are advised to go through the list carefully and take necessary action.

| | Documents | Enclosed Yes/No |
|---|---|---|
| 1) | Proof of Tender fee paid and duly notarized EMD declaration as per Appendix-9 of terms and conditions | |
| 2) | In case of tender fee exemption required, A letter stating exemption of Tender fee with copy of valid MSE (bidders registered with UDYAM REGISTRATION PORTAL)/NSIC/KVIB certificates | |
| 3) | Tender Index as per Appendix 1 of this RFP | |
| 4) | Power of Attorney or document as per clause 11,Chapter 2 of this tender document and as per Appendix 2 | |
| 5) | Copy of Partnership deed or Company Registration certificate or AoA/MoA or Other deed of constitution(in case of other entities) | |
| 6) | Schedule of deviation if any from Tender terms and conditions as per Appendix 4 | |
| 7) | Letter of proposal as per Appendix 5 of this tender document | |
| 8) | Documents pertaining to clause 14.a, Chapter 2 of this tender document | |
| 9) | Copy of CA audited balance sheet, profit and loss statement for financial year 2021-22, 22-23, and 2023-24 pertaining to clause 14. b chapter2 and format as per Appendix 11 and 11A of this tender document | |
| 10) | Documents pertaining to clause 14.c, Chapter 2 of this tender document and as per format Appendix 3 of this tender document | |
| 11) | Undertaking that applicant is not blacklisted /debarred as per clause 14.d Chapter 2 as per format Appendix 8 of this tender document | |
| 12) | Documents pertaining to clause 14.e, Chapter 2 of this tender document | |
| 13) | Documents pertaining to clause 14.f,Chapter 2 of this tender document | |
| 14) | Documents pertaining to clause 14.g, Chapter 2 of this tender document | |
| 15) | Duly filled company profile as per Appendix 7 of this tender document with necessary attachments requested | |
| 16) | All required documents as per Chapter 02, clause 12 of the tender document to be furnished along with the technical bid | |
| 17) | Any other documents relevant to this tender | |
| 18) | Price Bid(BOQ) | |
| 19) | Integrity pact as per Appendix 13 | |

Place :

Date :

Name & Designation of Authorized Person

Signature & Stamp :

:

# APPENDIX 2

## POWER OF ATTORNEY

(To be printed on Non-Judicial Stamp paper of appropriate value and duly notarized)

To know all men by these presents that, I, _____, holding the post of _____ and competent authority of _____ (the Bidder/Lead Member of the Joint Venture/Consortium) do hereby constitute, appoint, authorise and nominate ----------(Name & Designation) whose signature is attested herein below, to do all such acts, deeds and things necessary to the application in connection or incidental with the RFQ/ RFP/Tender No: _____, floated by M/s Kochi Metro Rail Limited for----------) including signing and submission of all the documents and providing necessary information/response to Kochi Metro Rail Limited and also to bid, negotiate and also to execute the contract, in case is the tender is awarded.

This Power of Attorney shall remain valid, binding and irrevocable until the completion of the tender or till the completion of the tenure of contract to be executed between (the Bidder/Joint Venture/Consortium) and Kochi Metro Rail Limited, if tender is awarded in favour of (the Bidder / JV/Consortium), whichever is applicable.

We hereby agrees to ratify all the acts, deeds and things lawfully done by the Attorney pursuant to this Power of Attorney and that all acts, deeds and things done by above mentioned Attorney shall always be deemed to have been done by us on behalf of ------(the Bidder/ Joint Venture/Consortium).

IN WITNESS THEREOF, WE ---- (the Bidder/ Lead Member of the Joint Venture/Consortium as previously mentioned) have executed these presents on this ___day of _____at_____.

**For (Bidder/ Lead Member)**

**Competent Authority**

*Accepted.*

*… .....(Signature)*

*.....Name & Address of the Attorney*

Specimen signature of the Attorney is attested

**For------ (Bidder/ Lead Member)**

**Competent Authority**

# APPENDIX 3

## SCHEDULE OF COMPLIANCE WITH TECHNICAL SPECIFICATION OF ITEMS

| ITEM DESCRIPTION | Complied Yes/No | Remarks |
|---|---|---|
| Chapter 3 : Technical specification and Scope of work | | |

NB: Leaving this appendix blank will be treated as Compliance to the scope of work and technical specifications

Place :
Date :
Name & Designation of Authorized Person
Signature :
Stamp

# APPENDIX 4

## SCHEDULE OF DEVIATION FROM TENDER TERMS AND CONDITIONS

| (Tender clause / section no) | DEVIATION IF ANY | REMARKS |
|---|---|---|
| | | |

NB: Leaving this appendix blank will be treated as no deviation to the tender terms and conditions.

Place :
Date :
Name & Designation of Authorized Person
Signature :

# APPENDIX 5

## LETTER OF PROPOSAL SUBMISSION

Bid for Supply, Installation, Commissioning, Training and CAMC of X- Ray Baggage Scanning system & Door Frame Metal Detector.

To,

CGM(O&M)

Kochi Metro Rail Ltd

Reference : Tender for Supply of _____(Bid for Supply, Installation, Commissioning, Training and CAMC of X- Ray Baggage Scanning system & Door Frame Metal Detector)

Dear Sir,

We hereby submit our Bid for Supply, Installation, Commissioning, Training and CAMC of X-Ray Baggage Scanning system & Door Frame Metal Detector.

We enclose a Online payment details Transaction No .....dated.....for Rs ---------- ), towards Earnest Money Deposit and Tender Fee

DECLARATION:I/We hereby confirm having read and understood the tender documents and corrigendum/addendum if any and the requirements of work/supply under this tender. The information furnished by us is true to the best of our knowledge and belief and nothing has been concealed there from. I/We Agree to abide by all the terms and conditions of the tender documents.

Place :

Date :

Name & Designation of Authorized Person

Signature :

Stamp

# APPENDIX 6

## BANK GUARANTEE

**(for performance security)**

(To be printed on Non-judicial stamp paper of appropriate value)

This Deed of Guarantee made this the --- day of 20XX by (Name of the Bank), having its registered office at ----------, and one of its branches at----------(Address of the Branch) (hereinafter called the Guarantor) in favour of Kochi Metro Rail Ltd, ---------- (hereinafter called the beneficiary).

WHEREAS M/s. _____ (Name of the Contractor) having their address/ registered Office at _____(Address of the Contractor's registered Office) (hereinafter called the "Contractor")was awarded contract for (supply / Erection / Supply & Erection / Work/others- specify the purpose) of _____ (Name of the material / equipment / work/others- specify) by Kochi Metro Rail Limited , the "Beneficiary" and

WHEREAS a performance guarantee of Rs ------/- has to be submitted by the Contractor, before undertaking the contract and

WHEREAS the Contractor has requested the Guarantor for issuing a Bank Guarantee for Rs. _____(Amount as stipulated) valid till _____(mention here date of validity of this Guarantee which will be ------- days beyond the contract period) towards Performance guarantee amount payable to the Beneficiary, and

WHEREAS the Guarantor has agreed to issue such Bank Guarantee to the Beneficiary as hereunder mentioned:

We, _____ (Name of the Bank and address of the Branch giving the Bank Guarantee) having our registered Office at _____ (Address of Bank's registered Office) hereby give this Bank Guarantee No. _____ dated _____and do hereby irrevocably undertake to pay immediately on demand, through our designated Branch in Kochi, without requiring any previous notice and without any demur, reservation, recourse, contest or protest and without referring to any other sources including the Contractor and without the beneficiary having to substantiate its demand, to the beneficiary a sum not exceeding Rs._____ (amount as stipulated) (Rupees _____) (in words) on behalf of the Contractor. Guarantor agrees that any demand in writing made by the authorised officials of the Beneficiary shall be conclusive as regards the amount due and payable by the Guarantor under this Guarantee.

We, _____(Name of the Bank) further undertake to pay without demur the aforesaid amount in lump sum on demand or such part there of as the beneficiary may demand from time to time irrespective of the fact whether the said contractor admits or denies such claim or questions correctness in any court, Tribunal or Arbitration proceedings or before any authority. The aforesaid guarantee will remain in force and we shall be liable under the same irrespective of any concession or time being granted by the beneficiary to the contractor and this guarantee will remain in full force irrespective of any change of terms, conditions or stipulation or any variation in the terms of the said contract.

We,_____(Name of the Bank) further agree with the beneficiary that the beneficiary will have the fullest liberty without our consent and without in any manner affecting our obligations hereunder to vary any of the terms and conditions of the contract or to extend the time for performance of the contract by the contractor or to postpone any of the powers exercisable by the beneficiary or to forbear or enforce any of the terms and conditions of the contract and that we shall not be relieved from our liability by reasons of any such variation or extension or forbearance or postponement or omission or by any indulgence by the beneficiary to the contractor or by any such matter whatsoever which under the law relating to sureties would, but for this provision, have resulted in relieving us.

This Bank Guarantee shall be valid and binding on this Bank upto and inclusive of _____(mention here the date of validity of Bank Guarantee), unless extended on demand by the beneficiary. The Guarantee shall not be terminated or affected by liquidation or winding up or insolvency or change in constitution of the Contractor or for any other reason. This guarantee shall not be terminated by the guarantor under any circumstances including change in the constitution of the Bank and our liability hereunder shall not be impaired or discharged by any extension of time or variations or alterations made, given, conceded in the contract with or without our knowledge or with or without consent by or between the Contractor and the beneficiary.

NOT WITHSTANDING anything contained hereinbefore, our liability under this Guarantee is restricted to Rs. _____ (amount as stipulated) (Rupees _____ (in words). Our Guarantee shall remain inforce till _____(  days after the date of validity of the contract). Unless demands or claims under this Bank Guarantee are made to us in writing on or before _____(date should be at least one year over and above the validity period of BG), all rights of Beneficiary under this Bank Guarantee shall be forfeited and we shall be released and discharged from all liabilities there under.

**Place:**

**Signature of the Bank's Authorized**

**Date:** **Signatory with Official Seal**

# APPENDIX 7

## COMPANY PROFILE

| No. | DETAILS | |
|---|---|---|
| 1 | Title | Company/Firm/Individual/HUF/Partnership/Others(Others-Please specify) |
| 2 | Name of Bidder<br>(with city and postal code) | |
| 3 | Address with telephone, e-mail, fax number & the name(s) of the contact person(s) | |
| 4 | GST ID<br>State of registration<br>(attach GST<br>Registration certificate) | |
| 5 | PAN DETAILS (attach documentary proof duly attested) | |
| 6 | BANK DETAILS<br>(Attach cheque copy)<br>Name of Bank<br>Name of Branch<br>IFSC Code<br>Account Number<br>Address of Bank | |

NB:Duly filled company profile along with cheque copy,PAN card copy and GST certificate to be scanned and uploaded as a single document

Place :
Date :
Name & Designation of Authorized Person
Signature :
Stamp :

# APPENDIX 8

## LETTER OF UNDERTAKING

To,

CGM (O&M)

Kochi Metro Rail Ltd

This is to certify that M/S…….....a reputed and certified Company/firm/Sole proprietorship firm for Supply, Installation, Commissioning, Training and CAMC of X- Ray Baggage Scanning system & Door Frame Metal Detector has NOT been Black listed /Debarred by any Government undertaking /semi Government/ Govt. Controlled Institutions /projects /Reputed private organizations in India or Abroad.

If it is found at any stage of the tender processing or after the award of the contract that, we are Blacklisted/Debarred by any organization as mentioned hereinabove, KMRL reserve the absolute right to take appropriate action against us, including but not limited to the termination of the award of tender or initiating legal proceedings against us.

Place :
Date :

Name & Designation of Authorized Person

Signature :

Stamp

# APPENDIX 9

## UNDERTAKING FOR TENDER SECURITY DECLARATION

### BID SECURING DECLARATION

(To be printed on Non-judicial stamp paper of Rs.500/-)

To,

Kochi Metro Rail Limited,
4th Floor, JLN Stadium Metro Station,
Banerjee Road, Kaloor, Ernakulam 682017

Ref: Tender Document No: _____dated _____

I/We, _____(hereinafter referred as bidder), declare that:

The bidder understand that, according to terms and conditions of the bid document as referred herein above, bids must be supported by a Bid Securing Declaration.

The bidder understands and accepts that it may be disqualified from bidding in any tender/quotation with Kochi Metro Rail Limited for a period of one year from the date of notification if, the bidder is in breach of any of the following obligation(s) under the bid conditions,

a. if the bidder resile or withdraw its proposal/tender after the bid closing date/ last date of submission of tender; OR

b. if the bidder do not accept the correction of Tender price during evaluation.

The bidder understands and accepts that it may be disqualified from bidding in any tender/quotation with Kochi Metro Rail Limited for a period of two years from the date of notification if the bidder is in breach of any obligation under the bid conditions,

I. if, having been notified of the acceptance of its bid/tender by KMRL during the period of tender validity, the bidder

II. fail or refuse to furnish the unconditional acceptance of Contract order within the time limit specified in bid document.

III. fail or refuse to furnish the Security deposit/ Performance Security in accordance with bid document .

For the purpose of this Bid Securing Declaration, the period of disqualification shall be from the date of resiling / withdrawal of this tender or from date of issue of LOA/WO/PO/GeM Contract order or from the date of such other breach, as the case may be.

The bidder understand this Bid Securing Declaration shall cease to be valid if the bidder is not the successful Bidder, upon the earlier of

I. the receipt of your notification of the name of the successful Bidder; or

II. Thirty days after the expiration of the validity of the bid.

For _____

**Signature**

**Name of authorised signatory:**

**Designation:**

**(Office Seal)**

# APPENDIX-10

## AGREEMENT

This agreement is executed on this the ----- day of ------, 20XX

between

**Kochi Metro Rail Ltd**., a Joint Venture Company of Government of India & Government of Kerala, incorporated under the Companies Act, 1956, for setting up of Metro Rail Project at Kochi, having its registered Office at 4th Floor, JLN Metro Station, Kaloor, Kochi – 682017 [hereinafter called **"the Employer/KMRL"** which the expression shall unless otherwise repugnant to the context shall means and include all its successors and permitted assigns], represented by its , Shri ---------- on the FIRST PART

**and**

----------, ----------, ---------- hereinafter called **"the Contractor" " [**which the expression shall unless otherwise repugnant to the context shall means and include all its successors and permitted assigns], represented by its--------, Shri---------- on the OTHER PART.

For the purpose of this agreement both Employer and contractor are collectively called 'Parties' and in individually called 'Party'.

WHEREAS the employer has floated a Tender vide No: ---------- dated ---------- for the "----------" (herein after referred as "Work") and the contractor turned out- to be the successful bidder, and

WHEREAS the employer has issued the Letter of Award No ---------- dated - ---------- to the contractor in compliance with the tender conditions and whereas the parties have decided to reduce their understanding into writing.

**NOW THIS AGREEMENT WITNESSETH AS FOLLOWS:**

**Clause 1 Words and Expression**

In this Contract, all the words and expressions shall have the same meanings as are respectively assigned to them in the conditions of contract in the Tender documents hereinafter referred to

**Clause 2 The Contract Documents,**

The following documents shall be deemed part of this agreement.

1. The Letter of Award (LOA).
2. The Letter of Bid and Appendix to Bid (including the signed Statement of Integrity)
3. The Particular Conditions
4. The General Conditions;
5. The Technical Specifications
6. The Drawings; and
7. The completed Schedules and any other documents forming part of the contract including tender documents,

All of the foregoing documents, together with this Agreement, are referred to herein as the Contract. Also incorporated into the Contract, and made part hereof, are all codes, standard specifications, and similar requirements that are referred to therein. In case of any discrepancy or ambiguity with documents mentioned above, this agreement shall have priority over all other documents.

**Clause 3 Obligations of the Contractor:**

That the Contractor agrees to perform efficiently and faithfully all of the Works related to ----------which is described in the Tender Notification No: ---------- and those works requisite for or incidental to the successful completion of the Works and in carrying out all duties and obligations imposed by the Contract.

That the Contractor agrees to confirm to the provisions of all statutes relating to the work and the regulations of local authorities concerned. The Contractor agrees to give all notices required by relevant Statutes or regulations and to pay all fees and taxes payable to any authority in respect thereof. The Contractor further agrees to comply with or cause to be complied with following statutory provisions and rules in respect of Personnel employed by them directly or through Sub-Contractors.

That the Contractor agrees to follow all the relevant statutes religiously and in particular the provisions of a) The Contract Labour (Regulation & Abolition) Act 1970 and the related Rules, b) The minimum wages Act 1948 and the related rules c) The payment of wages Act 1936 and the related rules, d) The Employees' Provident Fund & Miscellaneous provisions Act 1952, e) The Employees State Insurance Act 1948, f) Employees Compensation Act 1923 g) Payment of Bonus Act 1965 h) The Building and other Construction Workers Welfare Cess Act. 1996 i) Maternity

Benefit Act, 1961 j) Payment of Gratuity Act, 1972 k) Inter-State Migrant Workmen (Regulation of Employment & Conditions of Service) Act, 1979, l) Equal Remuneration Act, 1976 m) The Industrial Disputes Act 1947, n) Child Labour (Prohibition and Regulation) Act, 1986 o) Petroleum Act, 1934 and its related Rules p) Motor Vehicles Act, 1989 and its Rules q) Batteries (Management and Handling) Rules, 2001 r) Indian Electricity Rules, 1956 s) Environment (Protection) Rules, 1986 t) Hazardous Waste Management Rules, 2016 u) Solid Waste Management Rules, 2016 v) E-Waste(Management & Handling Rules), 2016, w) Gas Cylinder Rules, 2016 etc. and other statutes in general.

That the Contractor agrees to ensure the license for engaging labour is obtained from the appropriate government under the relevant Act/s and is live during the contractual period, and to have their own PF and ESI Codes and also to comply with the relevant Acts. Contractor agrees to maintain proper records of PF, EDLI, Pension, ESI contribution, administrative charges etc., wherever applicable and shall produce proof of deductions as well as remittances regularly. Contractor shall issue wage slips to his employees and shall maintain relevant registers and forms for inspection by various statutory authorities and the company Officials even at short notice. Contractor agrees to furnish proper Returns to the concerned statutory authorities and provide a copy of the same to KMRL.

That the Contractor agrees that it shall be solely responsible for non-payment / delayed payment of wages / DA, Bonus, contributions under EPF & MP Act, ESI Act etc. In case, the contractor fails to make payment of wages to his employees or remittance of contribution to the concerned authorities & a claim is made against KMRL for what so ever reason, the performance guarantee /other dues/ running bills under the contract can be utilized by KMRL to discharge such liability of the contractor, and the Contractor shall be liable to pay KMRL such amount thus paid. Contractor shall also be responsible for complying with provisions of Income Tax, Service Tax etc. as applicable to him.

**Clause 4 Obligations of the Employer:**

The Employer agrees, subject to the terms and conditions of the Contract, to pay the Contractor the amount specified, and at the rates and terms and in the manner set forth in the Contract.

**Clause 5 Contract Price and Completion Time:**

The Employer agrees to pay for the total cost of the Works and the Contractor agrees to accept the sums mentioned below in the following currencies, to be the total cost for the Work carried out by him as part of his obligations, responsibilities and liabilities under and according to the provisions and obligations imposed on him by the Contract.

Total Amount for Priced Bill of Quantities in all sections: -

(i) Indian Rupees ---------- (**Rs ----------/-**); including the GST

The Contractor shall complete whole of the Works within ---------- months from----------.

The Defect Liability Period is ---------months from the date mentioned in the Taking Over Certificate for whole of the Works. The actual payment of contract price will be depending on the actual works completed as per Bill of Quantities.

In consideration of the payments to be made by the Employer to the Contractor as specified in this Agreement, the Contractor hereby covenants with the Employer to execute the Works and to remedy defects therein in conformity in all respects with the provisions of the Contract.

The contractor has paid the performance security as per tender conditions vide ---------- for an amount of Rs ----------/- (Rupees ----------Only).

**Clause 6 Integration**

The Employer and the Contractor agree that this agreement, together with the other Contract Documents, expresses all of the agreements, understandings, promises, and covenants of the parties, and integrates, combines, and supersedes all prior and contemporaneous negotiations, understandings, and agreements, whether written or oral and no modification or alteration of the Contract Documents shall be valid or binding on either party, unless expressed in writing and executed with the same formality as this Agreement, except as may otherwise be specifically provided in the Contract Documents.

**Clause 7 Governing Law**

This Contract is enforceable and construed under the applicable laws of the Republic of India.

**Clause 8 Indemnity**

The Contractor shall indemnify KMRL against all claims and losses if it suffers under various labour laws, statutes or any civil or criminal law in connection with employees deployed by him or the work undertaken by him. The liability for any compensation on account of injury sustained by any employee of the contractor will be exclusively that of the contractor. KMRL shall not be responsible for any losses, damages to the contractor or to his employees. The contractor shall have full control over his employees including the right to appoint, determine service conditions,

discipline, discharge, dismissal etc. The Contractor shall be solely responsible for any claim arising out of employment or termination of employment of his employees and for their statutory payments. Contractor shall also ensure that workmen follow all rules and regulations related to safety and security. The Contractor shall abide by all the labour and other laws applicable to Contract labour / worker under this Contract and shall at all time keep KMRL Indemnified against all losses, claims, prosecutions under any law.

**Clause 9 Arbitration**

In case of any dispute or differences arising out of this agreement then the same shall be settled amicably between the parties by way of mutual discussion or negotiation. If such discussion or negotiation cannot resolve the issue within 30 days from the date of commencement of such discussion or negotiation then on the option of any of the parties, the same may be referred to a Sole Arbitrator to be appointed from the Panel of Arbitrators maintained by KMRL. If any dispute is raised by the contractor and arbitration clause is invoked, the Employer shall within 30 days of such invocation, submit to the contractor a panel having not less than 4 Arbitrators, from which the Contractor shall chose one as Sole Arbitrator. Arbitration shall be conducted under the provisions of the Arbitration and conciliation Act, 1996. Venue and seat of the arbitration shall be Ernakulam. Language of the proceedings shall be English. Award of the sole arbitration shall be final and binding upon the parties.

**Clause 10 Jurisdiction of Court**

Subject to above mentioned arbitration clause, any dispute or differences arising out of this agreement shall fall under the exclusive jurisdiction of court at Ernakulum.

**Clause 11**

All other terms and conditions mentioned in the other documents forming part of the contract shall be complied with by the parties.

**IN WITNESS WHEREOF,** the parties hereto have caused this Contract to be signed their respective names as of the day and year first above written.

| | |
|---|---|
| **For Kochi Metro Rail Limited** | **For Contractor** |
| **Authorised Signatory** | **Authorised Signatory** |

Witnesses

# APPENDIX 11

## Financial DATA

(Ref. Clause.14.b chapter 2 of this RFP)

Applicant's legal name ……

Date …….....

Page …….. of Pages

Each Applicant must fill in this form

| S.No. | Description | Financial Data for Latest Last 3 Years (Indian Rupees) | | |
|---|---|---|---|---|
| | | 2021-22 | 2022-23 | 2023-24 |
| 1. | Gross Annual turnover | | | |

Attach copies of the audited balance sheets, including all related notes, income statements for the last three audited financial years, as indicated above, complying with the following conditions.

1. Historic financial statements shall be audited by Statutory Auditor of the Company under their seal & stamp and shall be strictly based on Audited Annual Financial results of the relevant period(s). No statements for partial periods will be accepted.
.

2. This Appendix shall be duly certified by Chartered Accountant / Company Auditor under his signature, stamp and membership number.

# APPENDIX 11A

{Affidavit for Unaudited Balance Sheet as per clause 14.b chapter 2 of this RFP}

(On a non-judicial stamp paper of appropriate value)

(To be filled by the bidder* in case their Balance Sheet for F.Y. 2023-24 has not been audited. If the Balance sheet for F.Y. 2023-24 has been audited then the bidder need not to fill this form or may simply write "NOT APPLICABLE")

I,…….…….. (Name and designation of Authorised sSgnatory) of (Name of Company/ Firm/ Proprietorship/ Partnership) hereby confirm that the Balance sheet for Financial year i.e. F.Y. 2023-24 has actually not been audited/ or under finalization so far.

Signature of authorized signatory on behalf of Tenderer

# APPENDIX 12

**EMD BANK GUARANTEE**

(To be printed on Non-judicial stamp paper of appropriate value)

This Deed of Guarantee made this the --- day of 20XX by -------- (Name of the Bank), having its registered office at ----------, and one of its branches at -------- (Address of the Branch) (hereinafter called the Guarantor) in favour of Kochi Metro Rail Ltd, ---------- ----------(hereinafter called the beneficiary).

WHEREAS M/s. _____(Name and Address of the Entity participating in the tender) having their address/ registered Office at _____(Address of the Firm's registered Office) (hereinafter called the "Tenderer") wish to participate in the Tender No. _____for _____of (supply / Erection / Supply & Erection / Work/others- specify the purpose) of _____(Name of the material / equipment / work/others- specify) floated by Kochi Metro Rail Limited (hereinafter called the "Beneficiary") and

WHEREAS an EMD of Rs. -----/- has to be submitted by the Tenderer for participating in the aforesaid Tender and

WHEREAS the tenderer has requested the Guarantor for issuing a Bank Guarantee for Rs. _____(Amount of EMD) valid till _____(mention here date of validity of this Guarantee which will be---------- days beyond initial validity of Tender) towards EMD payable to the Beneficiary, and

WHEREAS the Guarantor has agreed to issue such Bank Guarantee to the Beneficiary as hereunder mentioned:

We, _____ (Name of the Bank and address of the Branch giving the Bank Guarantee) having our registered Office at _____ (Address of Bank's registered Office) hereby give this Bank Guarantee No. _____dated _____ and do hereby irrevocably undertake to pay immediately on demand, through our designated Branch in Kochi, without requiring any previous notice and without any demur, reservation, recourse, contest or protest and without referring to any other sources including the Tenderer and without the beneficiary having to substantiate its demand, to the beneficiary a sum not exceeding Rs._____ (amount of EMD) (Rupees _____) (in words) on behalf of the Tenderer. Guarantor agrees that any demand in writing made by the authorised officials of the Beneficiary shall be conclusive as regards the amount due and payable by the Guarantor under this Guarantee.

We, _____(Name of the Bank) further undertake to pay without demur the aforesaid amount in lump sum on demand or such part there of as the beneficiary may demand from time to time irrespective of the fact whether the said tenderer admits or denies such claim or questions correctness in any court, Tribunal or Arbitration proceedings or before any authority. The aforesaid guarantee will remain in force and we shall be liable under the same irrespective of any concession or time being granted by the beneficiary to the tenderer and this guarantee will remain in full force irrespective of any change of terms, conditions or stipulation or any variation in the terms of the said tender.

This Bank Guarantee shall be valid and binding on this Bank upto and inclusive of _____(mention here the date of validity of Bank Guarantee) and shall not be terminated or affected by liquidation or winding up or insolvency or change in constitution of the tenderer or for any other reason. This guarantee shall not be terminated by the guarantor under any circumstances including change in the constitution of the Bank and our liability hereunder shall not be impaired or discharged by any extension of time or variations or alterations made, given, conceded with or without our knowledge or with or without consent by or between the Tenderer and the beneficiary.

NOT WITHSTANDING anything contained hereinbefore, our liability under this Guarantee is restricted to Rs. _____ (amount of EMD) (Rupees _____ (in words). Our Guarantee shall remain inforce till _____( days after the date of validity of the Bid/tender as stipulated). Unless demands or claims under this Bank Guarantee are made to us in writing on or before _____ ((date should be at least one year over and above the validity period of BG), all rights of Beneficiary under this Bank Guarantee shall be forfeited and we shall be released and discharged from all liabilities there under.

**Place:**

**Signature of the Bank's Authorized**

**Date:** **Signatory with Official Seal**

# APPENDIX 13

## INTEGRITY PACT

Between

Kochi Metro Rail Limited (KMRL) hereinafter referred to as "The Principal",
And ---------- hereinafter referred to as **"The Bidder/ Contractor"**

**Preamble.**

The Principal intends to award, under laid down organizational procedures, contract/s for---------- The Principal values full compliance with all relevant laws of the land, rules, regulations, economic use of resources and of fairness / transparency in its relations with its Bidder(s) and / or Contractor(s).

In order to achieve these goals, the Principal will appoint Independent External Monitors (IEMs) who will monitor the tender process and the execution of the contract for compliance with the principles mentioned above.

**Section 1 - Commitments of the Principal**

1) The Principal commits itself to take all measures necessary to prevent corruption and to observe the following principles: -

a) No employee of the Principal, personally or through family members, will in connection with the tender for, or the execution of a contract, demand, take a promise for or accept, for self or third person, any material or immaterial benefit which the person is not legally entitled to.

b) The Principal will, during the tender process treat all Bidder(s) with equity and reason. The Principal will in particular, before and during the tender process, provide to all Bidder(s) the same information and will not provide to any Bidder(s) confidential / additional information through which the Bidder(s) could obtain an advantage in relation to the tender process or the contract execution.

c) The Principal will exclude from the process all known prejudiced persons.

2) If the Principal obtains information on the conduct of any of its employees which is a criminal offence under the IPC/PC Act, or if there be a substantive suspicion in this regard, the Principal will inform the Chief Vigilance Officer and in addition can initiate disciplinary actions.

**Section 2 - Commitments of the Bidder(s)/ Contractor(s)**

1) The Bidder(s)/ Contractor(s) commit themselves to take all measures necessary to prevent corruption. The Bidder(s)/ Contractor(s) commit themselves to observe the following principles during participation in the tender process and during the contract execution.

a) The Bidder(s)/ Contractor(s) will not, directly or through any other person or firm, offer, promise or give to any of the Principal's employees involved in the tender process or the execution of the contract or to any third person any material or other benefit which he/she is not legally entitled to, in order to obtain in exchange any advantage of any kind whatsoever during the tender process or during the execution of the contract.

b) The Bidder(s)/ Contractor(s) will not enter with other Bidders info any undisclosed agreement or understanding, whether formal or informal. This applies in particular to prices, specifications,

certifications, subsidiary contracts, submission or non-submission of bids or any other actions to restrict competitiveness or to introduce cartelisation in the bidding process.

c) The Bidder(s)/ Contractor(s) will not commit any offence under the relevant IPC/PC Act; further the Bidder(s)/ Contractor(s) will not use improperly, for purposes of competition or personal gain, or pass on to others, any information or document provided by the Principal as part of the business relationship, regarding plans, technical proposals and business details, including information contained or transmitted electronically.

d) The Bidder(s)/ Contractor(s) will, when presenting their bid, disclose any and all payments made, is committed to or intends to make to agents, brokers or any other intermediaries in connection with the award of the contract.

e) Bidder(s) /Contractor(s) who have signed the Integrity Pact shall not approach the Courts while representing the matter to IEMs and shall wait for their decision in the matter.

2) The Bidder(s)/ Contractor(s) will not instigate third persons to commit offences outlined above or

or be an accessory to such offences.

**Section 3 - Disqualification from tender process and exclusion from future contracts**

If the Bidder(s)/Contractor(s), before award or during execution has committed a transgression through a violation of Section 2, above or in any other form such as to put their reliability or credibility in question, the Principal is entitled to disqualify the Bidder(s)/Contractor(s) from the tender process or take action.

**Section 4 - Compensation for Damages**

1) If the Principal has disqualified the Bidder(s) from the tender process prior to the award according to Section 3, the Principal is entitled to demand and recover the damages equivalent to Earnest Money Deposit/ Bid Security.

2) If the Principal has terminated the contract according to Section 3, or if the Principal is entitled to terminate the contract according to Section 3, the Principal shall be entitled to demand and recover from the Contractor liquidated damages of the Contract value or the amount equivalent to Performance Bank Guarantee.

**Section 5 - Previous transgression**

1) The Bidder declares that no previous transgressions occurred in the last three years with any other Company in any country conforming to the anti-corruption approach or with any Public Sector Enterprise in India that could justify his exclusion from the tender process.

2) If the Bidder makes incorrect statement on this subject, he can be disqualified from the tender process or action can be taken as per the procedure mentioned in “Guidelines on Banning of business dealings”.

**Section 6 - Equal treatment of all Bidders / Contractors / Subcontractors**

1) In case of Sub-contracting, the Principal Contractor shall take the responsibility of the adoption of Integrity Pact by the Sub-contractor.

2) The Principal will enter into agreements with identical conditions as this one with all Bidders and Contractors.

3) The Principal will disqualify from the tender process all bidders who do not sign this Pact or violate its provisions.

**Section 7 - Criminal charges against violating Bidder(s) / Contractor(s) / Subcontractor(s)**

If the Principal obtains knowledge of conduct of a Bidder, Contractor or Subcontractor, or of an employee or a representative or an associate of a Bidder, Contractor or Subcontractor which constitutes corruption, or if the Principal has substantive suspicion in this regard, the Principal will inform the same to the Chief Vigilance Officer.

**Section 8 - Independent External Monitor**

1) The Principal appoints competent and credible Independent External Monitor for this Pact after approval by Central Vigilance Commission. The task of the Monitor is to review independently and objectively, whether and to what extent the parties comply with the obligations under this agreement.

2) The Monitor is not subject to instructions by the representatives of the parties and performs his/her functions neutrally and independently. The Monitor would have access to all Contract documents, whenever required. It will be obligatory for him / her to treat the information and documents of the Bidders/Contractors as confidential. He/ she reports to the management of KMRL.

3) The Bidder(s)/Contractor(s) accepts that the Monitor has the right to access without restriction to all Project documentation of the Principal including that provided by the Contractor. The Contractor will also grant the Monitor, upon his/her request and demonstration of a valid interest, unrestricted and unconditional access to their project documentation. The same is applicable to Sub-contractors.

4) The Monitor is under contractual obligation to treat the information and documents of the Bidder(s)/ Contractor(s) / Sub-contractor(s) with confidentiality. The Monitor has also signed declarations on 'Non-Disclosure of Confidential Information' and of 'Absence of Conflict of Interest'. In case of any conflict of interest arising at a later date, the IEM shall inform management of KMRL and recuse himself / herself from that case.

5) The Principal will provide to the Monitor sufficient information about all meetings among the parties related to the Project provided such meetings could have an impact on the contractual relations between the Principal and the Contractor. The parties offer to the Monitor the option to participate in such meetings.

6) As soon as the Monitor notices, or believes to notice, a violation of this agreement, he/she will so inform the Management of the Principal and request the Management to discontinue or take corrective action, or to take other relevant action. The monitor can in this regard submit non-binding recommendations. Beyond this, the Monitor has no right to demand from the parties that they act in a specific manner, refrain from action or tolerate action.

7) The Monitor will submit a written report to the management of KMRL within 8 to 10 weeks from the date of reference or intimation to him by the Principal and, should the occasion arise, submit proposals for correcting problematic situations.

8) If the Monitor has reported to the management of KMRL, a substantiated suspicion of an offence under relevant IPC/ PC Act, and the management of KMRL has not, within the reasonable time taken visible action to proceed against such offence or reported it to the Chief Vigilance Officer, the Monitor may also transmit this information directly to the Central Vigilance Commissioner.

9) The word 'Monitor' would include both singular and plural.

**Section 9 - Pact Duration**

This Pact begins when both parties have legally signed it. It expires for the Contractor 12 months after the last payment under the contract, and for all other Bidders 6 months after the contract has been awarded. Any violation of the same would entail disqualification of the bidders and exclusion from future business dealings.

If any claim is made / lodged during this time, the same shall be binding and continue to be valid despite the lapse of this pact as specified above, unless it is discharged / determined by management of KMRL.

**Section 10 - Other provisions**

1) This agreement is subject to Indian Law. Place of performance and jurisdiction is the Registered Office of the Principal, i.e. New Delhi.

2) Changes and supplements as well as termination notices need to be made in writing. Side agreements have not been made.

3) If the Contractor is a partnership or a consortium, this agreement must be signed by all partners or consortium members.

4) Should one or several provisions of this agreement turn out to be invalid, the remainder of this agreement remains valid. In this case, the parties will strive to come to an agreement to their original intentions.

5) Issues like Warranty / Guarantee etc. shall be outside the purview of IEMs.

6) In the event of any contradiction between the Integrity Pact and its Appendix, the Clause in the Integrity Pact will prevail.

(For & On behalf of the Principal)

(Office Seal)

Place

Date

(For & On behalf of Bidder/ Contractor)

(Office Seal)

Witness 1:(Name & Address)

Witness 2:(Name & Address)

**……THE END…….**

**.....THE END.....**